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Total Plant Solution (TPS) System

Time Link and GPS Antenna User's Guide

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TotalPlant

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About This Document

References

The following list identifies all documents that may be sources of reference for material discussed in this publication.

Document Title	Doc ID
Time Link / GPS Antenna Troubleshooting Guide	TP80000
Interpret System Clock Functions	L61524 LCN

Contacts

World Wide Web

The following Honeywell web sites may be of interest to Industry Solution customers.

Honeywell Organization	WWW Address (URL)
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About This Document

Symbol Definitions

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Symbol Definitions

The following table lists those symbols used in this document to denote certain conditions.

Symbol	Definition
	ATTENTION: Identifies information that requires special consideration.
	TIP: Identifies advice or hints for the user, often in terms of performing a task.
	REFERENCE -EXTERNAL: Identifies an additional source of information outside of the bookset.
	REFERENCE - INTERNAL: Identifies an additional source of information within the bookset.
CAUTION	Indicates a situation which, if not avoided, may result in equipment or work (data) on the system being damaged or lost, or may result in the inability to properly operate the process.

Symbol	Definition
	CAUTION: Indicates a potentially hazardous situation that, if not avoided, may result in minor or moderate injury. It may also be used to alert against unsafe practices. CAUTION symbol on the equipment refers the user to the product manual for additional information. The symbol appears next to required information in the manual.
	WARNING: Indicates a potentially hazardous situation that, if not avoided, could result in serious injury or death. WARNING symbol on the equipment refers the user to the product manual for additional information. The symbol appears next to required information in the manual.
	WARNING, Risk of electrical shock: Potential shock hazard where HAZARDOUS LIVE voltages greater than 30 Vrms, 42.4 Vpeak, or 60 VDC may be accessible.
	ESD HAZARD: Danger of an electro-static discharge to which equipment may be sensitive. Observe precautions for handling electrostatic sensitive devices
	Protective Earth (PE) terminal: Provided for connection of the protective earth (green or green/yellow) supply system conductor.
	Functional earth terminal: Used for non-safety purposes such as noise immunity improvement. NOTE: This connection shall be bonded to Protective Earth at the source of supply in accordance with national local electrical code requirements.
	Earth Ground. Functional earth connection. NOTE: This connection shall be bonded to Protective Earth at the source of supply in accordance with national and local electrical code requirements.
	Chassis Ground: Identifies a connection to the chassis or frame of the equipment shall be bonded to Protective Earth at the source of supply in accordance with national and local electrical code requirements.

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1. System Timing Requirements vs. Clock Options

1.1 System Clock Overview

Overview

In some applications, it may be desirable to reference TPS system time to a stable time reference such as UTC (Universal Time Coordinated). In these situations, an option is available for GUS/APP nodes and TPN Bridge nodes to satisfy these requirements. The GPS antenna provides a stable system clock that is referenced to UTC (Universal Time Coordinated) by means of time message packets and a one pulse per second timing signal. In addition to a GPS time reference, it may be desirable to provide low time skew between LCN segments in a TPN Bridge application. In this situation, a time link cable may be routed between TPN Bridge nodes. Figure 1 below shows a TPN Bridge application in which both the GPS and Time Link options are utilized. Additional examples are presented in section "Time Link Cable and GPS Options" of this guide.

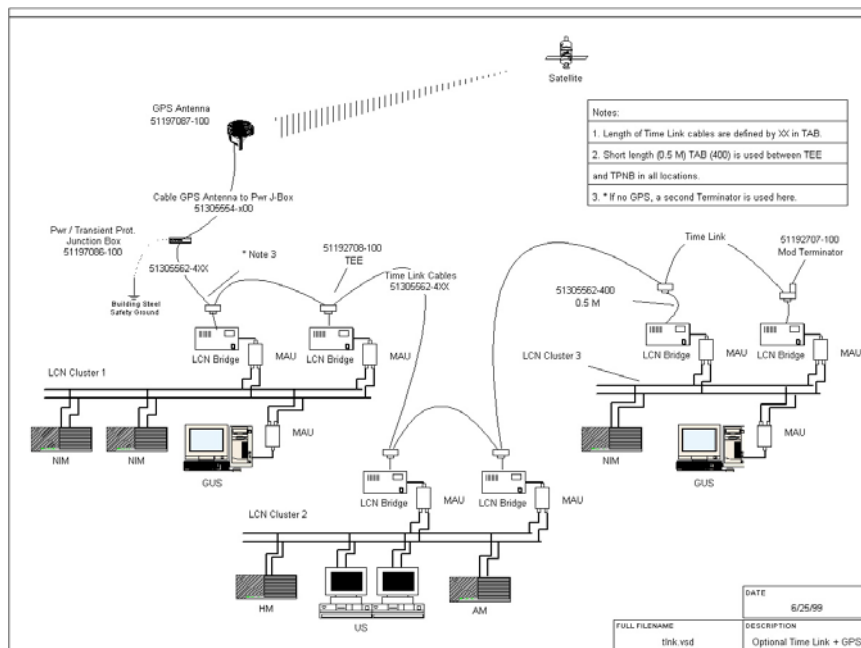


Figure 1-1 Utilization of GPS and Time Link in TPN Bridge

1. System Timing Requirements vs. Clock Options

1.2. Tasks for configuring the system clock

1.2 Tasks for configuring the system clock

Overview

The following tasks, which are detailed in this section, are associated with properly configuring the system for the desired timing attributes.

- Determining System Timing Requirements
- Understanding how the Sync Reference Options relate to System Timing Requirements
- Making the Time Link Master / Segment Master Configuration Selections
- Setting the TPN Master and Segment Master Configuration

Determining System Timing Requirements

The amount of time skew allowed between LCN segments is a function of the application. For TPN system applications that can tolerate up to 16 ms of segment-to-segment skew, no Time Link cabling is required.

If low ($\pm 300 \mu\text{s}$) skew between LCN segments is required, but long-term system time accuracy is not required, then the addition of the shielded twisted pair (STP) Time Link cable fills this need.

If the ± 16 ms segment to segment skew is acceptable, but long term accuracy of the system time is required, this requirement may be met by the connection of the GPS antenna to two Bridge partners (TPNBs) or to two GUS nodes with the LCNP4 assemblies that are GPS capable. This is particularly useful in systems without TPS ac reference capability (all PC based systems.)

If low ($\pm 300 \mu\text{s}$) skew between LCN segments is required and long term accuracy of the system time is required, then the GPS antenna and Time Link STP connection is extended to additional pairs of TPN Bridge nodes.

The system software (RNOS), which is running on the LCN nodes, has specific expectations in regards to calendar time ownership, clock mode and synchronization state management, and fail-over recovery. The TPN Bridge operates in a manner that allows the software on existing LCN nodes to operate without modification. When it is desirable to replace the existing LCN clock source node operation with a GPS reference, an NCF configuration menu change is required to set the clock source node addresses to the Bridge Node address assignments.

Understanding how the Sync Reference Options relate to System Timing Requirements

The following synchronization reference (source) options are supported:

- For the TPN Bridge, synchronization to existing (LCN) clock sources allows the segmentation of existing LCN systems to be performed without modification to the NCF (Network Configuration File) and the associated re-load of nodes to pick up the new NCF configuration.
- Synchronization to a GPS timing reference is an option that allows the TPN system time to track UTC. This may be accomplished in one of two ways:
 - GPS antenna connection to a redundant pair of TPN Bridge nodes, or
 - GPS antenna connection to GUS/APP nodes with GPS capable LCNP4s.

The nodes (GUS/APP or TPN Bridge) that are to act as the interface between the GPS antenna and the system **MUST** be configured in the NCF as clock sources. If the node addresses of the selected GUS/APP or TPN Bridge nodes are different than the current NCF assignments, then the NCF must be updated to show this new configuration. Only one set of clock sources is allowed per system. If GPS is used, the nodes that act as the interface between the GPS antenna and the system must be the clock sources.

1. System Timing Requirements vs. Clock Options

1.2. Tasks for configuring the system clock

As you can see, it is not a low segment-to-segment skew requirement that drives the re-configuration of the system clock sources in the NCF, but a system requirement for system time that has a deterministic relationship to UTC. The following table is a summary of how the system timing requirements drive the sync source (clock source) selection.

Table 1-1 Summary of Timing Requirements vs. Sync Source

System Timing Requirement		Existing LCN Based Clk Sources	GPS Connection to TPN Bridges	Seg-to-Seg Time Link Cabling
UTC Reference	Seg-to-Seg Skew			
No	+/- 16 ms	Okay	No	No
No	+/- 300 μ s	Okay	No	Yes (Note 1)
Yes	+/- 16 ms	No, CLK Sources must be GPS Capable.	Either TPN Bridge (2 Redundant Partners)	No
Yes	+/- 300 μ s	No, CLK Sources must be GPS Capable.	Either TPN Bridge (2 Redundant Partners)	Yes (Note 1)
Note 1: A Time Link cable may be connected only to TPN Bridge nodes or to GUS/APP nodes. Time Link mastership rules prevent the connection of TPN Bridge nodes and GUS/APP nodes to the same Time Link cable.				

Making the Time Link Master / Segment Master Configuration Selections

TPN Bridge Specific Information

In the previous section information was presented to aid in understanding how the Sync (timing source) reference options relate to the system timing requirements. When it comes to time messaging it is important to know how the TPN Bridge nodes know "Who is the Boss." This section will discuss how to make the selections based upon the results of the previous section.

1. System Timing Requirements vs. Clock Options

1.2. Tasks for configuring the system clock



ATTENTION

A GPS cable is defined as the cable between the antenna and junction box. A Time Link cable is defined as the cable(s) from the junction box to the nodes.

All bridge nodes automatically determine the synchronization reference to be used by detecting the assertion of specific signals within the GPS and Time Link cables. On a node-by-node basis, the three “system level synchronization references” indicate if that particular node is to get its sync reference from the GPS and Time Link cables, from the Received Time Frame mailbox or from neither. The following table defines the terms and definitions for the TPN Master and Segment Master configuration on each TPN Bridge.

Table 1-2 **Defining the TPN and Segment Master**

Term	Definition
<i>TPN Master Role (TLMr)</i>	This is the node currently responsible for distribution of time data to all TPN Segments.
<i>Segment Master Role (SMr)</i>	This is the node currently responsible for transmission of time data to the connected (one) Segment.
<i>Primary/Secondary</i>	A Bridge state associated with Open LAN operation and activation of specific time synchronization Roles based upon configuration bits TLMc and TSMc.

1. System Timing Requirements vs. Clock Options

1.2. Tasks for configuring the system clock

Setting the TPN Master and Segment Master Configuration

TPN Bridge Specific Information

The user is required to know how to set the TPN and Segment Master configuration for each TPN Bridge. The purpose of this section is to provide instruction on the correct settings of the TPN Master and Segment Master configuration selection bits based upon the results of subsections, "Determining System Timing Requirements" and "Understanding how the Sync Reference Options relate to System Timing Requirements."

The following table associates the clock source selection for a three-segment system with the proper state of the TPN and Segment Master bits for each Bridge Pair. Redundant partners must always be configured in the same manner in regards to the TPN and Segment Master bits.

Table 1-3 Clock Source Association for Three-Segment System

Location of CLK Sources		Clock TPN Master / Segment Master Configuration Bits					
		Segment 1		Segment 2		Segment 3 ...	
Segment # (1,2,3)	Type Bridge Pair / Not Bridge Pair	TPN Master	Segment Master	TPN Master	Segment Master	TPN Master	Segment Master
1	Not Bridge	Yes	No	No	Yes	No	Yes
1	Bridge	Yes	Yes	No	Yes	No	Yes
2	Not Bridge	No	Yes	Yes	No	No	Yes
2	Bridge	No	Yes	Yes	Yes	No	Yes
3	Not Bridge	No	Yes	No	Yes	Yes	No
3	Bridge	No	Yes	No	Yes	Yes	Yes

1.3 Time Link Cable and GPS Options and Examples

Scope

A Time Link cable option may be utilized to provide low skew between LCN segments.
A GPS antenna option may be utilized to provide accurate system time relative to UTC.

As shown in the following table, there are four options in Time Synchronization Accuracy in a Bridge Application.

Table 1-4 GPS Antenna Options for Accurate System Time Relative to UTC

Option	Description
1	No cable connections in terms of time link cabling. There is no GPS connection. This option only uses the software for segment synchronization. This is the simplest configuration and the least accurate (+/- 16 ms max skew between segments – No UTC reference for system).
2	Using a time link cable between the segments that provides low skew between segments. There is no reference to a GPS or Universal Time Clock. A low skew case but no reference to universal time. (+/- 300 μ s max skew between segments – No UTC reference for system).
3	Use of GPS connected to bridges on a single segment acting as a reference to Universal Time Clock but still distributing the time to other segments using the software. This provides good system time accuracy but also a fairly high segment-to-segment skew (+/- 16 ms max skew between segments – System is referenced to UTC).
4	Use of both GPS connection and time link cable connection to all segments. This provides low skew segment to segment and an accurate system time relative to UTC. (+/- 300 μ s max skew between segments – System is referenced to UTC.)



CAUTION

GPS time synchronization signals can only be daisy-chained between TPN Bridges or LCNs that are in the same building. A separate GPS system is required for LCNs located in separate buildings due to the differences in ground potentials between buildings.

1. System Timing Requirements vs. Clock Options

1.3. Time Link Cable and GPS Options and Examples

TPN Bridge – Example 1:

A system has three segments. The system clocks are two NIM's on the number 1 segment. The Bridge pair on the number 1 segment is configured with TPN Master (Yes), Segment Master (No). The Bridge pair on the number 2 and 3 segments is configured with TPN Master (No), Segment Master (Yes).

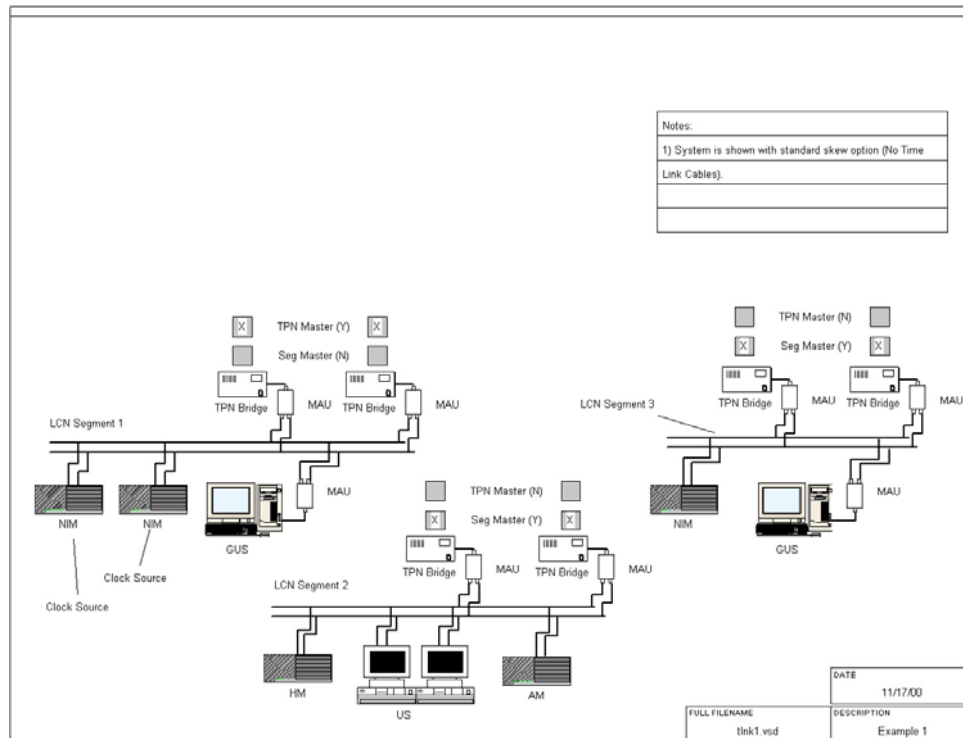


Figure 1-2 Configuration for Example 1

1.3. Time Link Cable and GPS Options and Examples

TPN Bridge – Example 2:

A system has three segments and utilizes Time Link cabling between all TPN Bridge nodes. The system clocks are two NIMs on the number 1 segment. The Bridge pair on the number 1 segment is configured with TPN Master (Yes), Segment Master (No). The Bridge pair on the number 2 and 3 segments is configured with TPN Master (No), Segment Master (Yes).

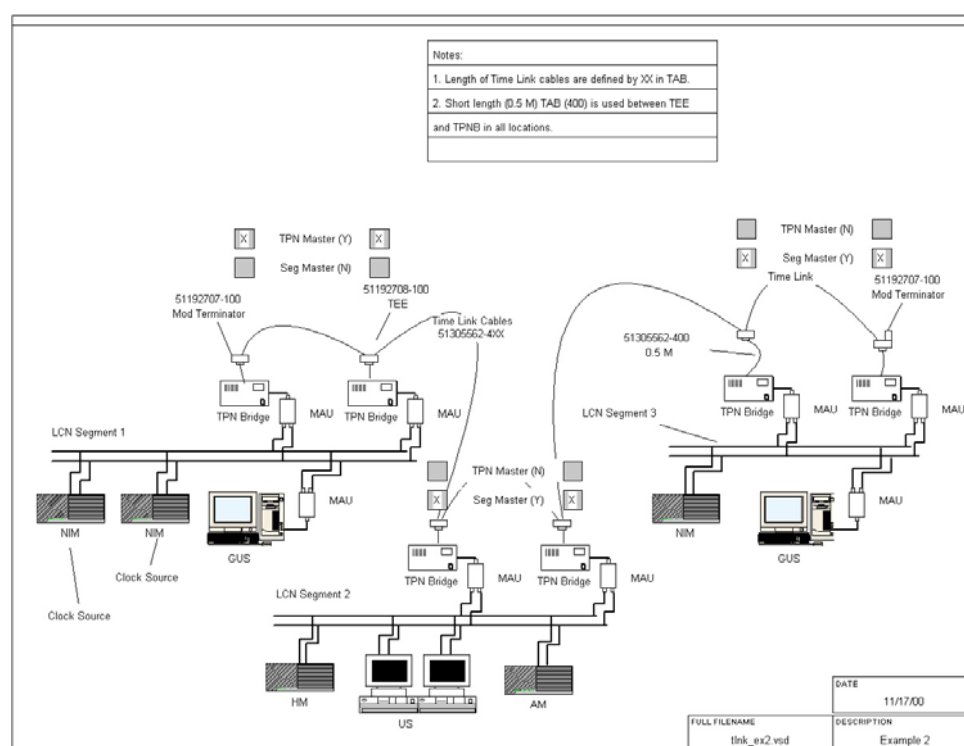


Figure 1-3 Configuration for Example 2

1. System Timing Requirements vs. Clock Options

1.3. Time Link Cable and GPS Options and Examples

TPN Bridge – Example 3:

A system has three segments and utilizes GPS with connections to two GUS nodes on the number 1 segment. The GUS nodes are configured as clock sources in the NCF. The Bridge pair on the number 1 segment is configured with TPN Master (Yes), Segment Master (No). The Bridge pair on the number 2 and 3 segments is configured with TPN Master (No), Segment Master (Yes).

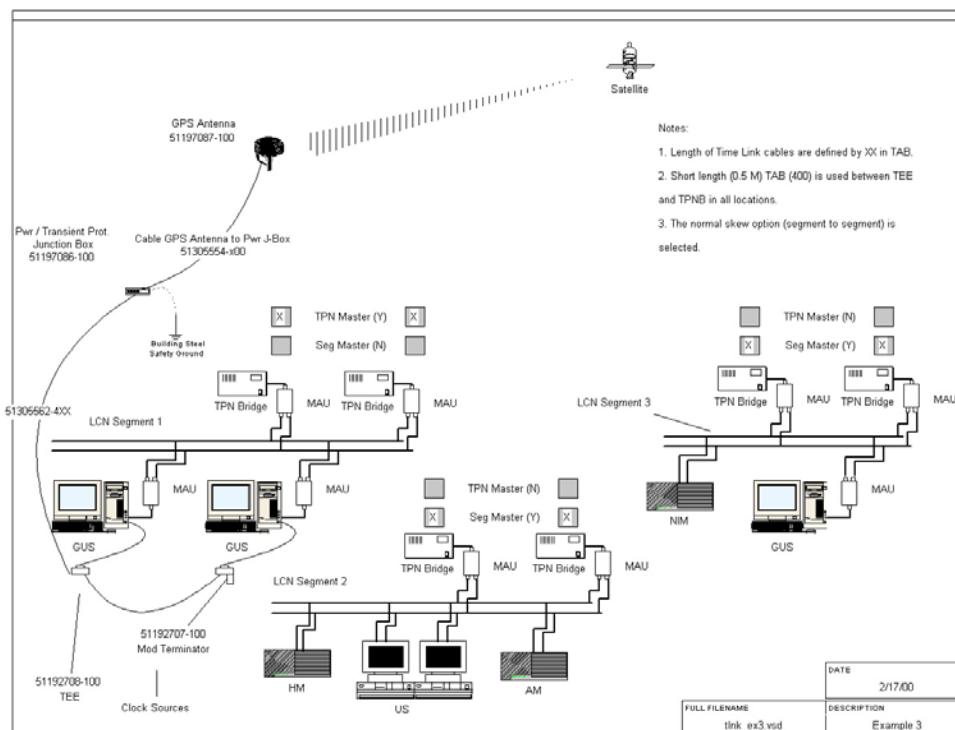


Figure 1-4 Configuration for Example 3

1. System Timing Requirements vs. Clock Options

1.3. Time Link Cable and GPS Options and Examples

TPN Bridge – Example 4:

A system has three segments and utilizes GPS with connections to the TPN Bridge partners on the number 2 segment. The NCF is configured with addresses 95 and 96 as the clock sources (TPN Bridge Nodes). The Bridge pair on the number 1 segment is configured with TPN Master (No), Segment Master (Yes). The Bridge pair on the number 2 segment is configured with TPN Master (Yes), Segment Master (Yes) and the Bridge pair on the number 3 segment is configured with TPN Master (No), and Segment Master (Yes).

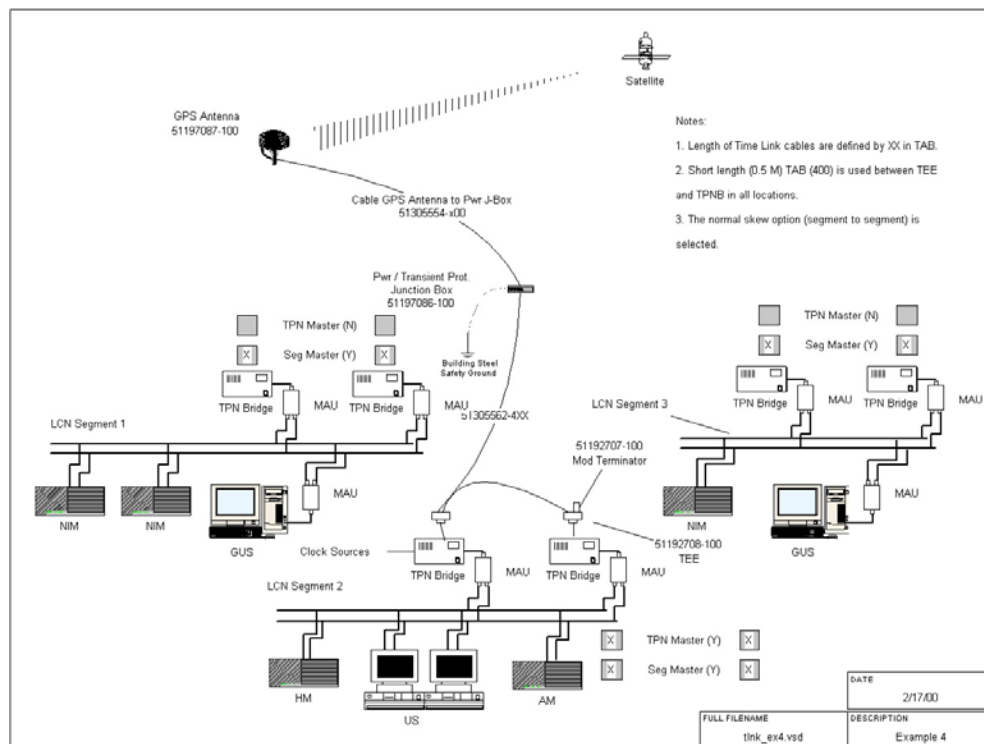


Figure 1-5 Configuration for Example 4

1. System Timing Requirements vs. Clock Options

1.3. Time Link Cable and GPS Options and Examples

TPN Bridge – Example 5:

This example is similar to Configuration for Example 2 except that a GPS signal reference has been added to the Time Link cable and the clock source nodes have been assigned to the TPN Bridge nodes on the LCN segment 1. In the event that there is a loss of the GPS signal, the LCN segments will be synchronized by means of the LAN interface (with the larger segment-to-segment skew) until the GPS signal is restored. During the period that the GPS signal reference is unavailable, the TPN Bridge nodes will periodically (every 4 minutes) attempt to re-establish sync to the GPS reference. Users will observe this sequence by monitoring the check boxes in the TDC status display of any TPN Bridge. The “Sync Queued,” “TL Port Disabled” will become checked followed by the “Am Synched” check box. When the re-sync (4 minute) timer has elapsed, an attempt to re-sync to the GPS signal will be performed. The “TL Port Disabled” will become unchecked and “Sync Queued” will become checked. If successful, the normal conditions shown in the table following the TDC Status Display figure will return (along with the flashing 1PPS LED). If not, the above sequence will continue to repeat. If the GPS signal has become permanently disabled for some reason and repairs are required, a terminator may be substituted for the Time Link cable that goes to the power junction box until a repair can be performed.

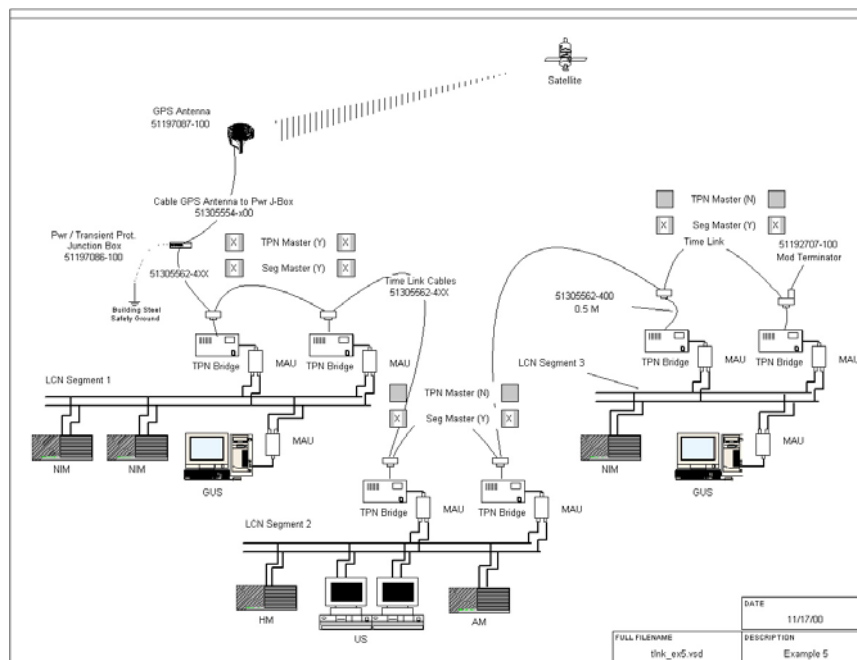


Figure 1-6 Configuration for Example 5

1. System Timing Requirements vs. Clock Options

1.3. Time Link Cable and GPS Options and Examples

Non-TPN Bridge – Example 6:

A system has a single LCN. A user wishes to have this LCN referenced to a common time source (GPS). In this example, the Time Link cable provides the GPS signal reference to a single clock source node. This option provides for continued system synchronization in the event of loss of the GPS system signal, but requires user intervention to force the Master Clock selection back to the GPS connected node as part of the recovery process.

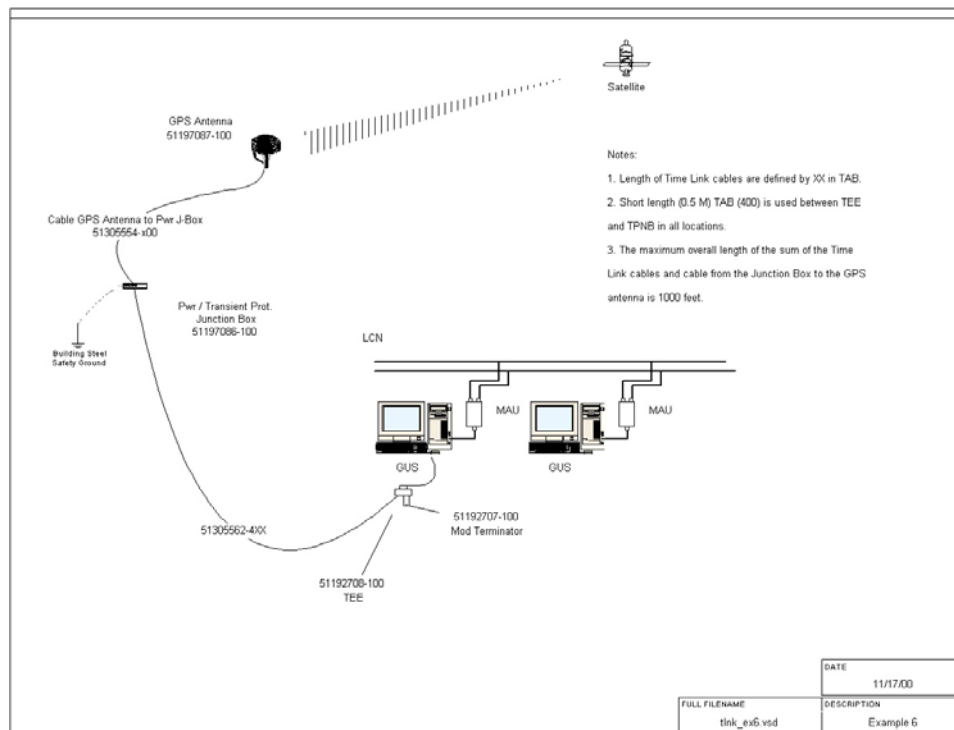


Figure 1-7 Configuration for Example 6

1. System Timing Requirements vs. Clock Options

1.3. Time Link Cable and GPS Options and Examples

Non-TPN Bridge – Example 7:

A system has a single LCN. A user wishes to have this LCN referenced to a common time source (GPS). In this example, the Time Link cable provides the GPS signal reference to a both clock source nodes. This option provides for a more automatic recovery process for the Master clock in the event of a short-term loss of GPS service. The disadvantage of this configuration with the current Clock Driver software is that both clock sources go to LOCAL mode while the GPS signal is unavailable. System synchronization may be restored if it is anticipated that the GPS signal will be unavailable for a long period of time. This is accomplished by disconnecting the Time Link cable to the far end clock source at the LCNP4 Time Link port jack. When GPS service is restored and the 1PPS LED flashing is visible (on the still connected LCNP4) then the connection can be restored to the second clock source and normal operation is restored.

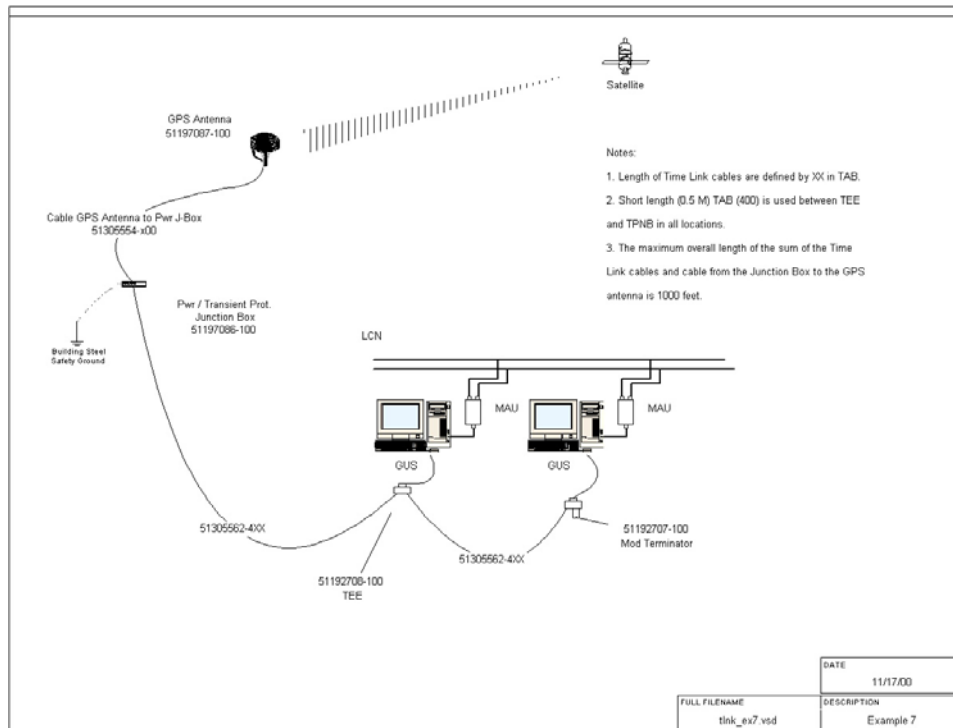


Figure 1-8 Configuration for Example 7

1. System Timing Requirements vs. Clock Options

1.3. Time Link Cable and GPS Options and Examples

Non-TPN Bridge – Example 8:

A system has 3 separate LCNs, which are independent from the perspective of LCN communications. A user wishes to configure all three LCNs referenced to a common time source (GPS). In this example, two separate Time Link cable systems, power junction boxes and GPS antennas are utilized to provide a redundant GPS signal referenced to each pair of clock source nodes on each LCN.

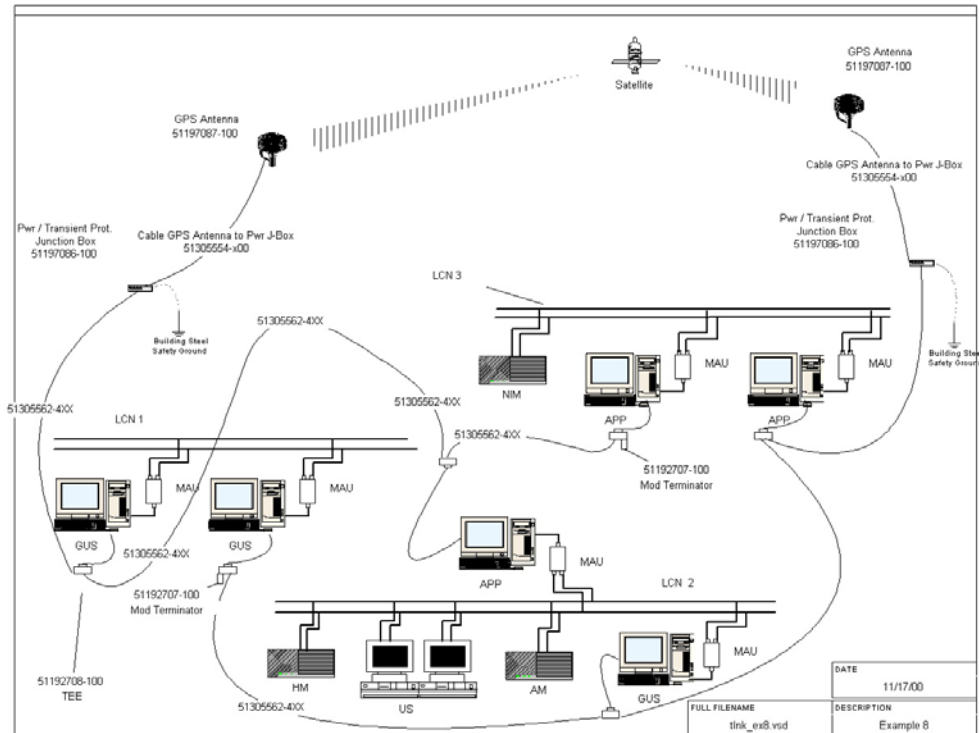


Figure 1-9 Configuration for Example 8

1. System Timing Requirements vs. Clock Options

1.3. Time Link Cable and GPS Options and Examples

2. Time Link (without GPS) Option for TPN Bridge

2.1 Overview

Purpose

This section specifically addresses the use of the **Time Link Option without GPS** in TPN Bridge applications. It is assumed that the user has read and understands the concepts associated with the following topics in Section 1.2 of this guide:

- Determining System Timing Requirements
- Understanding how the Sync Reference Options relate to System Timing Requirements
- Making the Time Link Master / Segment Master Configuration Selections
- Setting the TPN Master and Segment Master Configuration

It is also assumed that the user is familiar with Time Link elements (such as the Time Link Cables 51305562-4XX, TEE connectors 51192708-100, and Terminators 51192707-100).

2. Time Link (without GPS) Option for TPN Bridge

2.1. Overview

Time Link option without GPS reference

The following figure is an example of a TPN Bridge System, which uses the Time Link Option but does not use a GPS Reference.

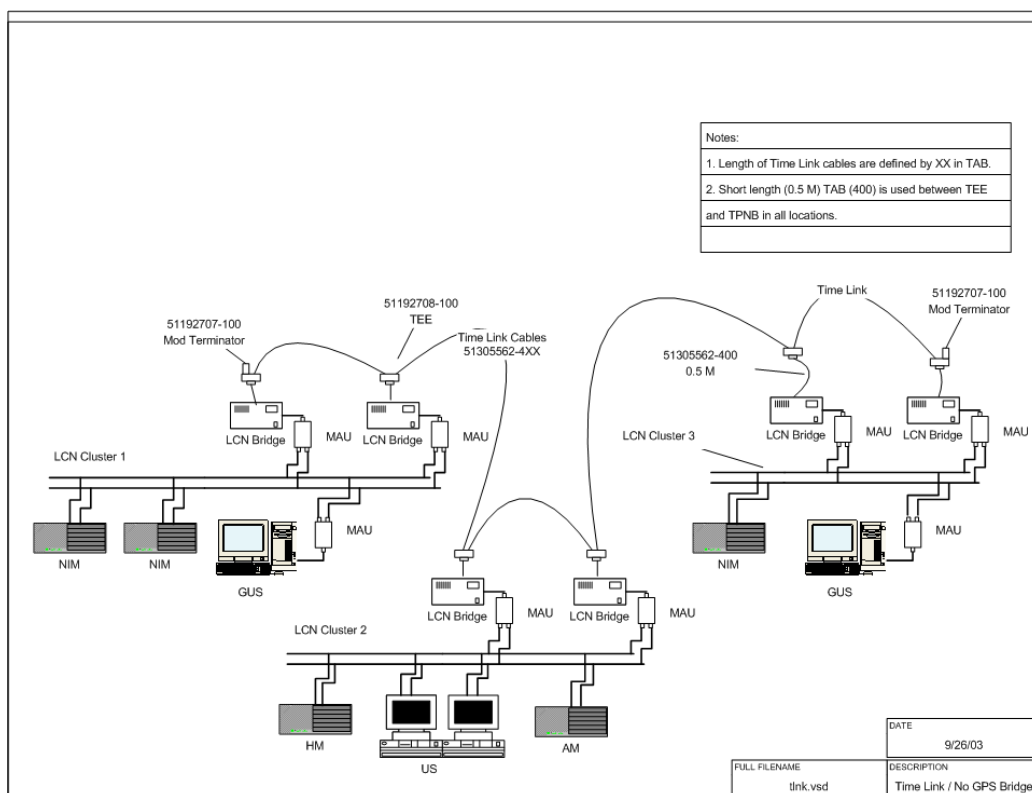


Figure 2-1 Time Link Option (without GPS) for TPN Bridge

Definitions

The following definitions apply to the topics and figures in this section.

Term	Definition
TPN Clock Master Primary/Secondary Bridges (bridge nodes with the TPN Master box checked in the "Time Mastership" section of the TDC Status display)	The primary bridge (and its redundant / secondary partner) on the bridge segment containing the LCN clock master and slave nodes. –OR– The primary bridge (and its redundant / secondary partner) that are selected to act as the LCN clock master and slave nodes AND TPN Clock Master nodes by means of the NCF and configuration (selection of both the TPN Master AND Segment Master boxes of the Time Mastership control) See the following subsections in Section 1.2 of this guide: "Making the Time Link Master / Segment Master Configuration Selections" "Setting the TPN Master and Segment Master Configuration" Note: Only one (1) bridge pair can be selected as TPN Clock Master.
TPN Clock non-Master Primary/Secondary Bridges (bridge nodes with the TPN Master box NOT checked in the "Time Mastership" section of the TDC Status display)	ALL OTHER BRIDGE PAIRS.

Assumptions

Figure 2-1 represents what a TPN Bridge system might look like **after** the installation of the Time Link components.

It is assumed, at this point, that the user has a TPN Bridge system that is currently operating without Time Link Components, and that the user wishes to install the Time Link cables, TEE connectors and Terminators.

It is also assumed that the Time Link cable length requirements have been identified, that all of the Time Link components have been ordered, and that they are ready for installation.

2. Time Link (without GPS) Option for TPN Bridge

2.2. Time Link Installation Flow

2.2 Time Link Installation Flow

Figure 2-2 and Figure 2-3 show the steps that should be used to ensure an orderly progression in time synchronization paths when performing the installation of Time Link cables. The step-by-step sequence also provides the user with improved ability to diagnose any problems to a specific operation or step. A brief text description of each identified step is provided in “Installation & Checkout” section of this document.

Time Link Cable Installation - Step A

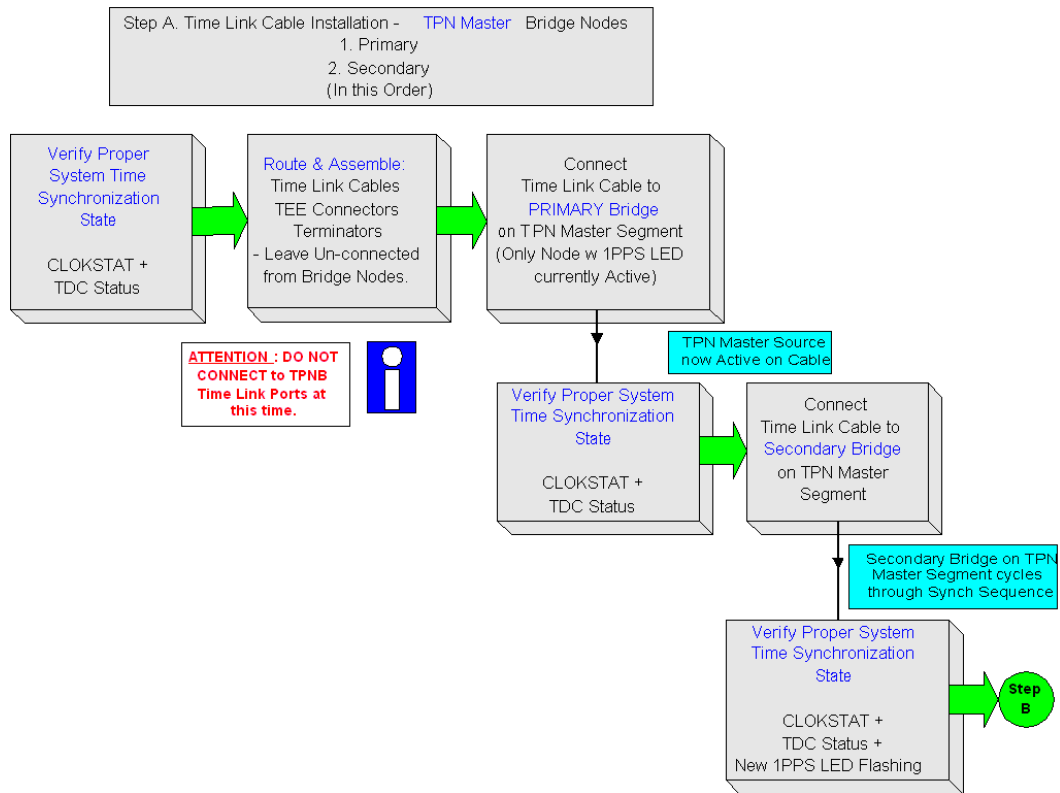


Figure 2-2 Time Link Cable Installation - step A

2. Time Link (without GPS) Option for TPN Bridge

2.2. Time Link Installation Flow

Time Link Cable Installation - Step B

Time Link cable connection to the PRIMARY Bridge on the TPN Master Segment should be performed first, followed by status check and connection to the SECONDARY Bridge on the TPN Master Segment. The Non-TPN Master Bridge nodes should follow in the sequence described below:

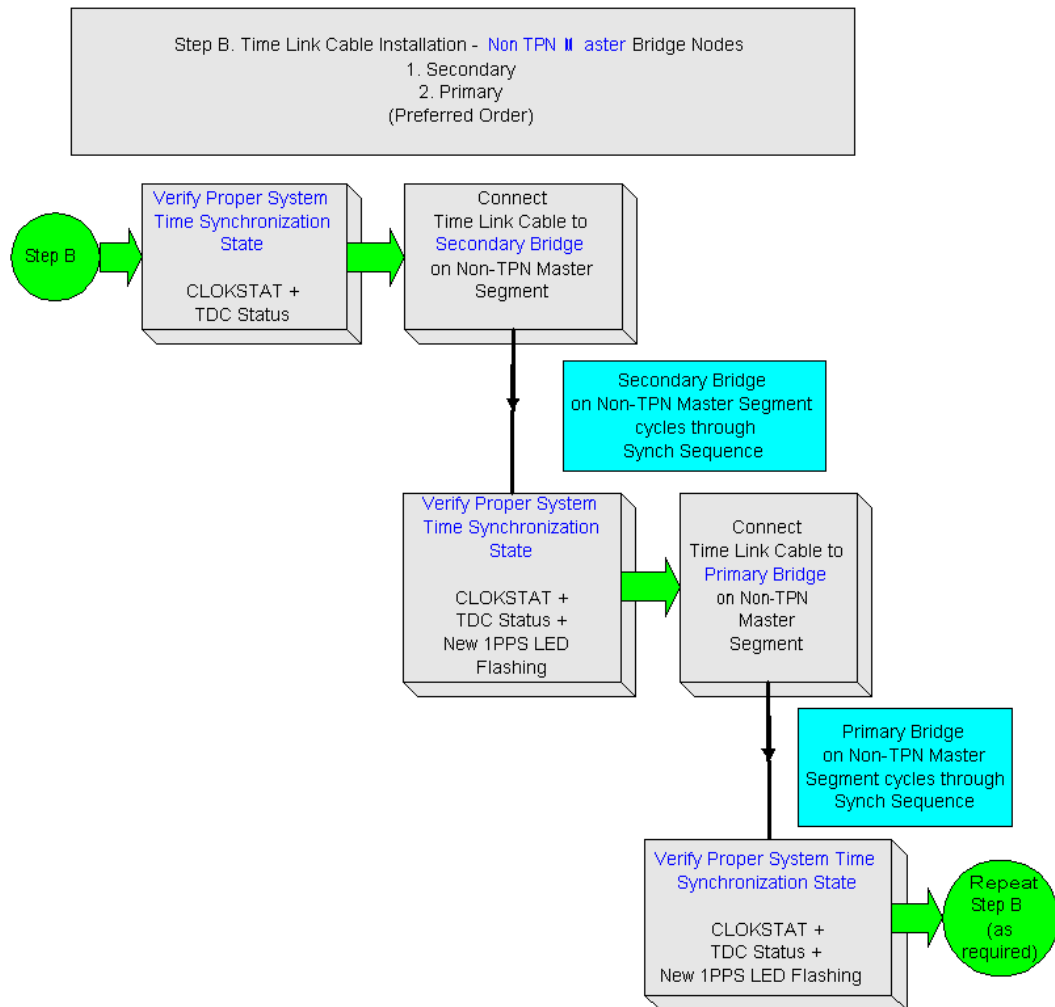


Figure 2-3 Time Link Cable Installation - step B

2. Time Link (without GPS) Option for TPN Bridge

2.2. Time Link Installation Flow

Time Link cable connection to Time Link port of a TPNB assembly

When a Time Link cable connection is made into the Time Link port of a TPNB assembly on a *Non-TPN Master* Segment, several things happen:

1. The 1PPS LED should start flashing. (Note: The TPN Master Primary Bridge Node is the only Node whose 1PPS LED should be flashing prior to the installation of the Time Link cable.)
2. The TPN Bridge Node performs a Sync sequence that can be observed with close observation of the TDC Status display. The sequence is demonstrated in Figure 2-5 and Figure 2-6.
3. Users should wait for the Sync sequence to complete before proceeding to the next Time Link cable connection.

The sequence described in Figure 2-3 should be repeated until all of the remaining Non-TPN Master Bridge nodes are connected.

2.3 TPN Bridge Status Displays

CLOKSTAT Schematic Display

The CLOKSTAT schematic, as shown in Figure 2-4, provides a good over-all view of the system time synchronization state. It is recommended that this or a similar status be monitored during the steps associated with the Time Link cable installation process. The following figure shows expected synchronization status for a representative LCN Bridge system. This particular system has 3 Bridge pairs with KxLCN type clock source nodes configured as node addresses 71 and 72. The KxLCN clock source nodes are located on Segment 2.

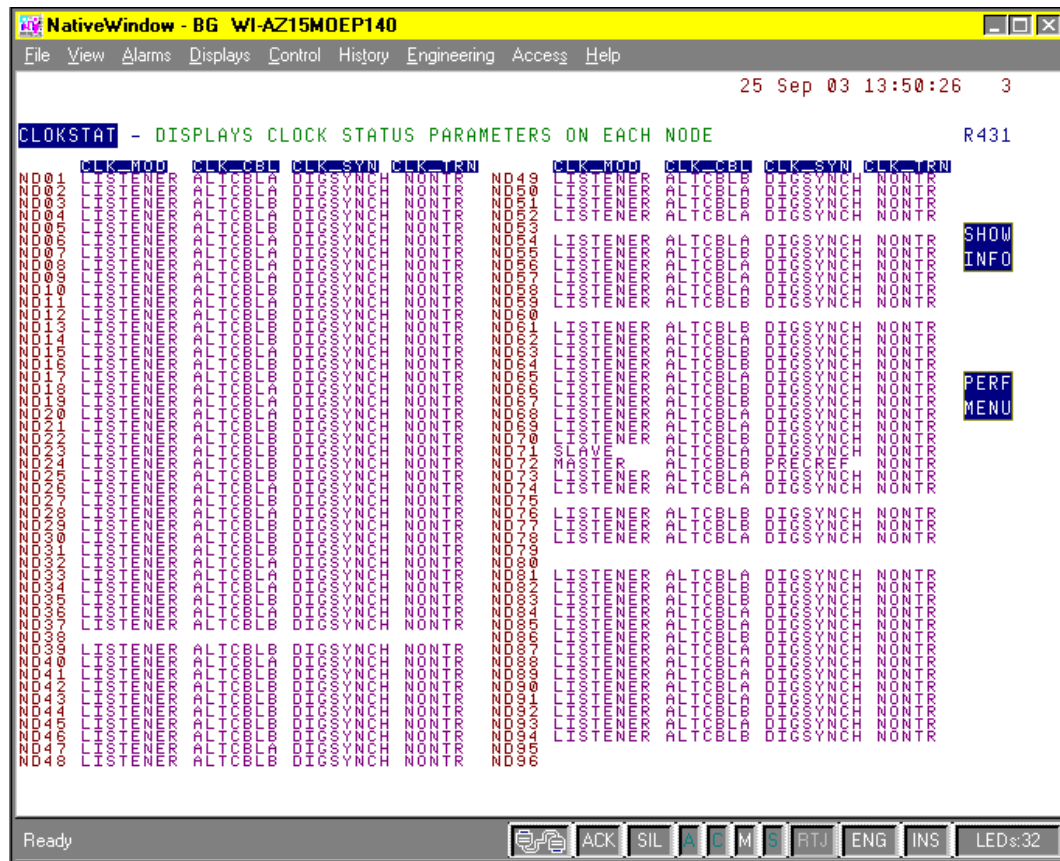


Figure 2-4 CLOKSTAT Schematic

2. Time Link (without GPS) Option for TPN Bridge

2.3. TPN Bridge Status Displays

Node addresses and segment location

A complete list of node addresses and Segment Location is shown in Table 2-1. Nodes 95 and 96 are the Bridge Node addresses.

Table 2-1 Example Node assignments / LCN Segment

Segment 1	Segment 2		Segment 3
Bridge Pair is Segment Master / Not-TPN Master	Bridge Pair is TPN Master / Not-Segment Master		Bridge Pair is Segment Master / Not-TPN Master
3	1	58	4
5	2	62	12
6	8	64	17
7	9	68	18
15	10	69	21
19	11	70	22
25	13	71	23
28	14	72	24
29	16	73	26
31	20	76	32
33	27	85	35
39	30	86	36
59	34	87	38
61	37	89	41
63	40	91	42
65	43	92	44
66	45	93	46
67	47	94	49
77	48		50
78	51		52
81	54		53
82	55		74
83	56		79
84	57		88
			90

TPN Master and Segment Master assignments

The TPN Master and Segment Master assignments are determined by the assignment of the clock source nodes. These assignments are defined in detail in the following subsections in Section 1.2 of this guide:

- Making the Time Link Master / Segment Master Configuration Selections
- Setting the TPN Master and Segment Master Configuration

2. Time Link (without GPS) Option for TPN Bridge

2.3. TPN Bridge Status Displays

Bridge Status and TDC Status Displays

Several TPN Bridge status displays are shown in the following figure. Of particular interest for recognition of the system time synchronization are the two fields circled below.

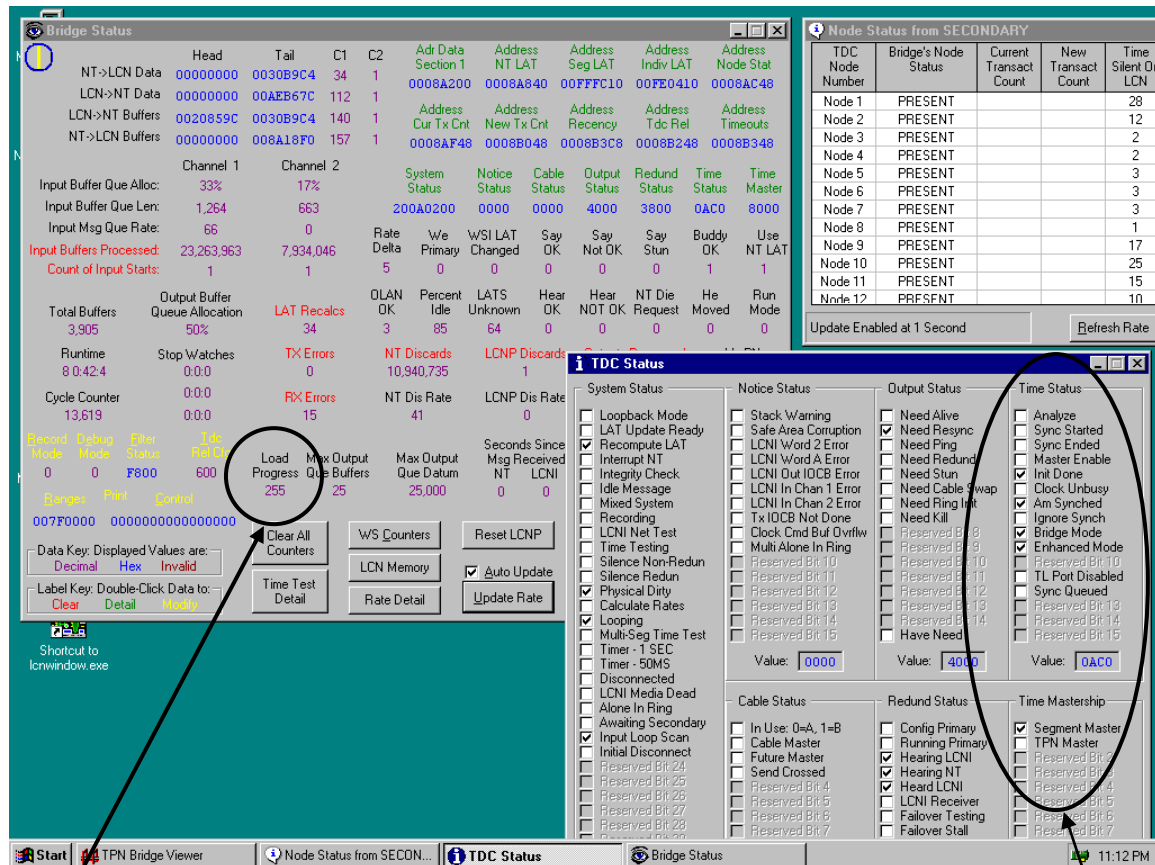


Figure 2-5

Bridge Status & TDC Status Displays

Load Progress of 255 Indicates Sync progress is complete.

Time Status indicates that Sync process is complete.
Time Mastership field indicates current TPN Master / Segment Master Selection for this TPN Bridge.

Time Link Cable Insertion Synchronization Cycle Sequence

When a Time Link cable connection is made into the Time Link port of a Non-TPN Master TPNB assembly (or TPN Master which is a Secondary Bridge) a sync state change occurs within the TPN Bridge node. This sync state change happens over the period of a small number of seconds from the time that the Time Link cable is inserted (or removed) into/from the Time Link port of the TPNB assembly. The following figure shows a typical change in synchronization state that may be observed (if a user is monitoring the TDC Status Display while a second person performs the Time Link cable insertion). This state change should last a few seconds followed by a return to the normal (steady state) sync state as indicated in Figure 2-5.

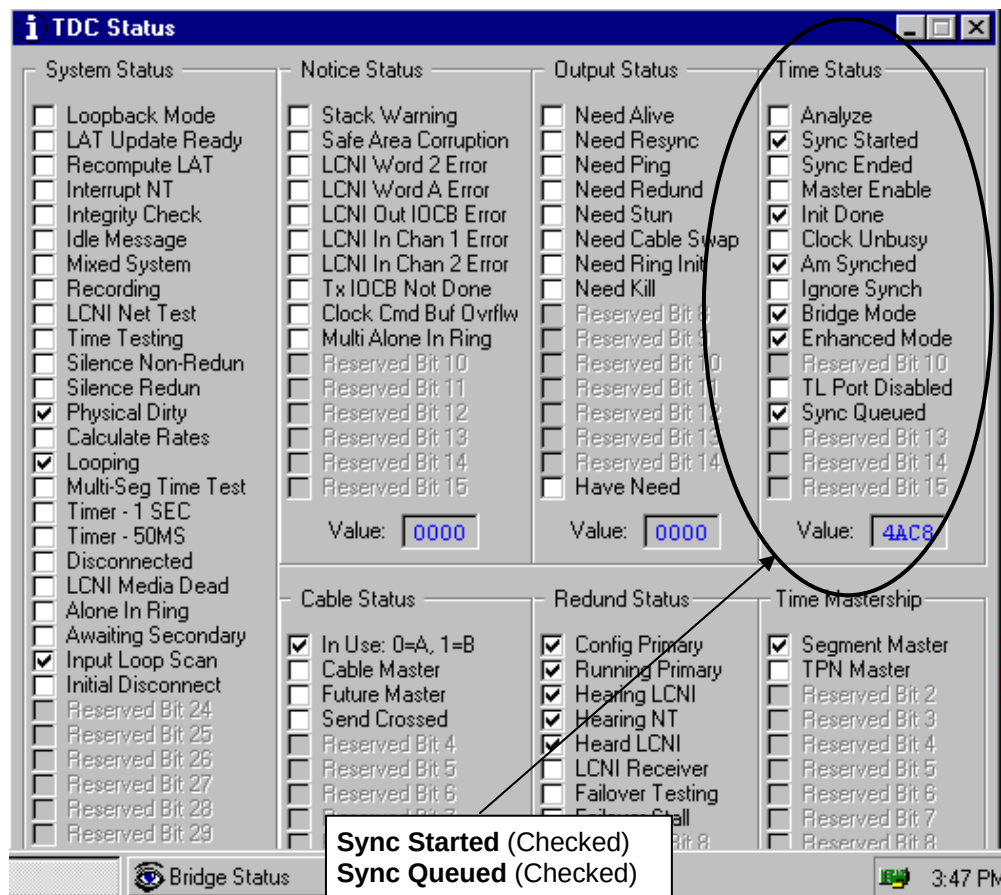


Figure 2-6 TDCStatus – Momentary Sync State Change

2.4 Installing Time Link Cable

Time Link Cable Installation and Checkout Action

The Time Link cable installation procedure follows:

Step	Installation Action	Done 3
1	Route and assemble the Time Link cables (51305582-4XX) into the TEE connectors (51192708-100) and install a Terminator at each end of the daisy chain (see Figure 2-1). DO NOT CONNECT the short 0.5M Time Link cables into the TPNB assemblies at this time.	☐
2	Before making Time Link cable connections to TPN Bridge nodes, verify that the System Time Synchronization is in a normal state using the CLOKSTAT schematic (see Figure 2-4) and verify that the current Time Status field and Load Progress field of the TDC Status display is as shown in Figure 2-5.	☐
3	Locate the Time Link Port (TLI) of the PRIMARY TPN Master Bridge node (it is the only TPNB that has an active 1PPS LED flashing each second). Locate a free end of one of the short 0.5M Time Link cables from the assembly performed in the Step 1 and connect this into the Time Link Port (TLI) of the PRIMARY TPN Master Bridge node.	☐
4	Verify that the System Time Synchronization remains in a normal state using the CLOKSTAT schematic (see Figure 2-4) and verify that the current Time Status field and Load Progress field of the TDC Status display is as shown in Figure 2-5.	☐
5	Locate the Time Link Port (TLI) of the SECONDARY TPN Master Bridge node. Locate a free end of one of the short 0.5M Time Link cables from the assembly performed in Step 1 and connect this into the Time Link Port (TLI) of the Secondary TPN Master Bridge node. Verify that the 1PPS LED becomes active when the cable is inserted into the Time Link port (TLI).	☐
6	Verify that the System Time Synchronization remains in a normal state using the CLOKSTAT schematic (see Figure 2-4) and verify that the current Time Status field and Load Progress field of the TDC Status display is as shown in Figure 2-5.	☐

NOTE: If a second person monitors the TDC Status display while the Time Link cable is being inserted, the second person should observe a **momentary sync state change** as indicated in Figure 2-6. This should only last for a small number of seconds and then the **status should return to the state shown in** Figure 2-5.

2. Time Link (without GPS) Option for TPN Bridge

2.4. Installing Time Link Cable

Step	Installation Action	Done 3
7	Locate the Time Link Port (TLI) of the SECONDARY Non-TPN Master Bridge node. Locate a free end of one of the short 0.5M Time Link cables from the assembly performed in step 1 and connect this into the Time Link Port (TLI) of the SECONDARY Non-TPN Master Bridge node. Verify that the 1PPS LED becomes active when the cable is inserted into the Time Link port (TLI).	☐
8	Repeat Step 6.	☐
9	Locate the Time Link Port (TLI) of the PRIMARY partner of the Non-TPN Master Bridge node that was connected in Step 7. Locate a free end of one of the short 0.5M Time Link cables from the assembly performed in Step 1 and connect this into the Time Link Port (TLI) of the PRIMARY Non-TPN Master Bridge node. Verify that the 1PPS LED becomes active when the cable is inserted into the Time Link port (TLI).	☐
10	Repeat Step 6.	☐
11	Repeat Steps 7 – 10 for each remaining Bridge Pair. It is recommended that the connections be performed on the Bridge pairs as indicated. If a problem is encountered, the cause of the problem should be identified and corrected before proceeding to the next step.	☐
Note: If it should become necessary to remove the Time Link cables, the removal of the cables should be performed in a reverse sequence from the installation of the cables.		

2. Time Link (without GPS) Option for TPN Bridge

2.4. Installing Time Link Cable

3. GPS / Time Link Component Descriptions

In TPS systems that do not have TPS Bridges, a GPS antenna may be connected to the GUS/APP node clock source nodes to provide accurate system time relative to UTC.



ATTENTION

Time Link cable connections between TPN Bridges nodes and GUS/APP node clock sources are not allowed. This is because there can only be one active Time Link Master allowed per Time Link cable and therefore only one pair of clock source nodes is allowed per Time Link Cable. The maximum overall length of all Time Link cables and the GPS cable is limited to 1000 feet.

The Time Link distribution cable consists of CAT5 STP cables, T-connectors and terminators.

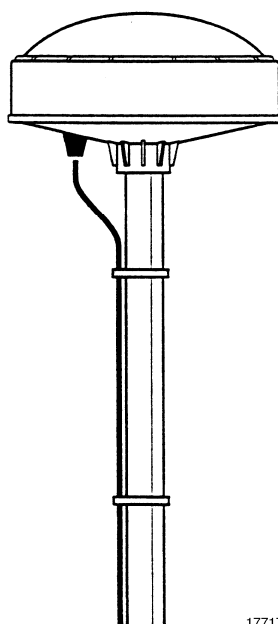
The GPS antenna option consists of the GPS antenna, power surge junction box, and cable between the antenna and the junction box.

3. GPS / Time Link Component Descriptions

3.1. Antenna

3.1 Antenna

The antenna, shown below, is a low profile, easy to mount device that must have line-of-site positioning to the satellites. The junction box can be located up to 61-meters (200 feet) from the antenna.



17717

Figure 3-1 Smart Antenna Mounted on a Standard 1" Marine & UNS Threaded Pipe Mast

Antenna placement

The antenna must be positioned with uninterrupted line of sight to the constellation of GPS satellites. This means that the area 30° above the horizon must be 100% clear.

Since the antenna itself contains active electronic components that could be damaged by direct lightning strikes, the antenna should be positioned with the intent of making it an unattractive path for lightning discharge. This may include the use of a suitable lightning rod discharge element with suitable ground wire. If a lightning rod is utilized, the GPS antenna should be positioned within the protective cone where the radius (r) of the base of the cone is equal to the height (h) of the top of the lightning rod.

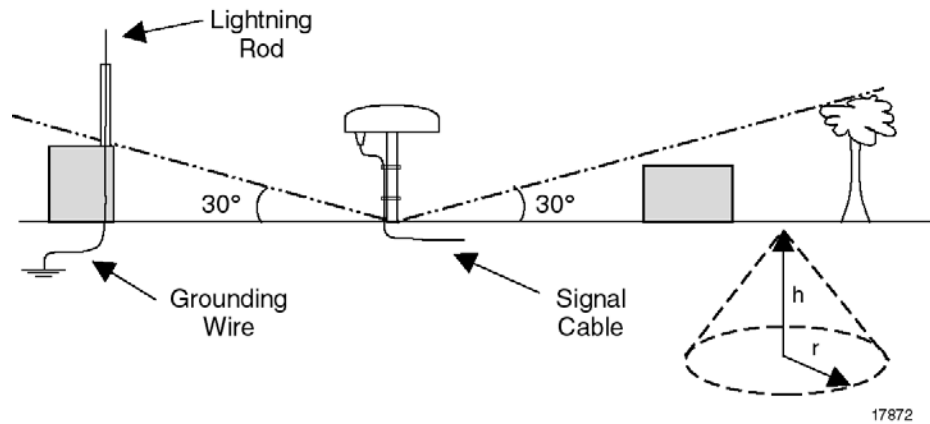


Figure 3-2 Positioning the GPS Antenna

3. GPS / Time Link Component Descriptions

3.2. GPS Cable, Antenna to Junction Box

3.2 GPS Cable, Antenna to Junction Box

The GPS cable, 51305554, is constructed with six twisted pair conductors, shielded and covered with UV stabilizer/inhibitor insulation. The cable comes in lengths of 15.2 meters (50 feet), 30.5 meters (100 feet) or 61 meters (200 feet).

Table 3-1 Antenna Cable Pin-Out

Wire Type	Signal RS422	Deutsch Pin #	Color	DB15 pin #
Paired	Power	1	RED	9
	Ground	9	BLACK	5 (note 1)
	———	—	BLACK	1 (note 1)
Paired	Port B Rcv -	2	WHITE	8
	Port B Rcv +	3	BLACK	11
Paired	Port B Tx -	4	YELLOW	3
	Port B Tx +	5	BLACK	10
Paired	Port A Rcv -	6	- - -	NC
	Port A Rcv +	7	- - -	NC
Paired	Port A Tx -	8	GREEN	2
	Port A Tx +	10	BLACK	7
Paired	1PPS +	11	BLUE	4
	1PPS -	12	BLACK	6
Drain Wires	NC	NC	Un-insulated	Shield

Note 1: This is a loopback wire for DB15 connector only.

3.3 Time Link Cables

Scope

There are two sets of redundant TPN Bridge LAN Ethernet cable, each identified by a colored plastic ring (boot) around each end of the cable. The 100-Base-T cables (CAT5 w/RJ45 Shielded Plugs) Time Link cable is identified with a color code of Black.

Description

Time Link cables bring the UTC signal from the junction box to the TPN Bridge or GUS/APP nodes. The use of these Time Link cables in the TPN Bridge system provides for low segment-to-segment skew. A ferrite bead inside adhesive shrink tubing at both ends uniquely identifies the Time Link cable.

The Time Link cable is available two ways: in standard lengths of 0.5, 2, 5, 10, 20, 50, 75, and 99 meters or custom specified lengths. All cables will have color-code of Black for identification boots (Tab 4XX where XX equals cable length) around each cable-end.

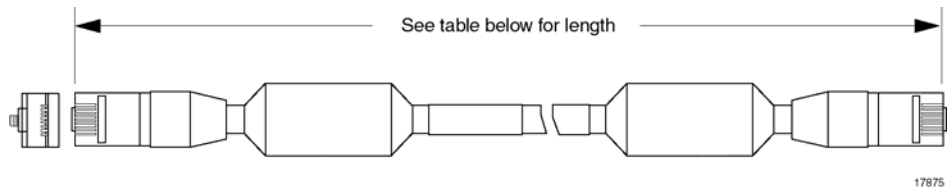


Figure 3-3 100 BaseTX Cable, 51305562-4XX

3. GPS / Time Link Component Descriptions

3.3. Time Link Cables

Standard Cable Lengths

Standard length cables are available by factory order.

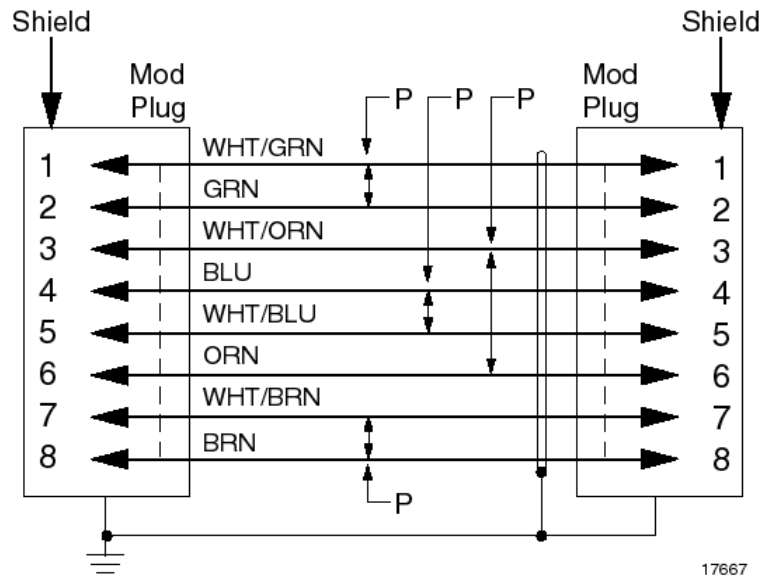


ATTENTION

The sum of all the Time Link cables and the GPS cable lengths is limited to 1000 feet.

**Table 3-2 Standard Cable Lengths for 100-Base-T Cable
w/Black Boots**

51305562-4XX Cable Tab	Standard Lengths
400	0.5 meters
402	2 meters
405	5 meters
410	10 meters
420	20 meters
450	50 meters
475	75 meters
499	99 meters
4XX	Custom Length XX meters

Cable construction**Figure 3-4 Cable Construction and Pin Assignments****Other Time Link Components****Table 3-3 Time Link T-Connector and Modular Terminator**

QTY	Description	Part Number
2	RJ45 (8x8) Modular Plug Terminator (100 Ohm)	51192707-100
AR	T Connector - Receptacle RJ45 - 2x8P8C to 8P8C Shielded	51192708-100

3.4 Power/Surge Junction Box

Does the following:

- provides DC power to the antenna receiver
- isolates the TPN system from power surges that may be induced in the antenna .

3. GPS / Time Link Component Descriptions

3.4. Power/Surge Junction Box

GPS Power Surge/Junction Box Cabling

The cabling connections to the Junction Box are shown in “Figure 11: GPS Power Surge/Junction Box Cabling Diagram.”



WARNING!

The antenna cable (to J2) shall be run through EMT or suitable other metal conduit and the conduit shall be suitably bonded to the GPS Power Junction Box by means of the mounting bracket.

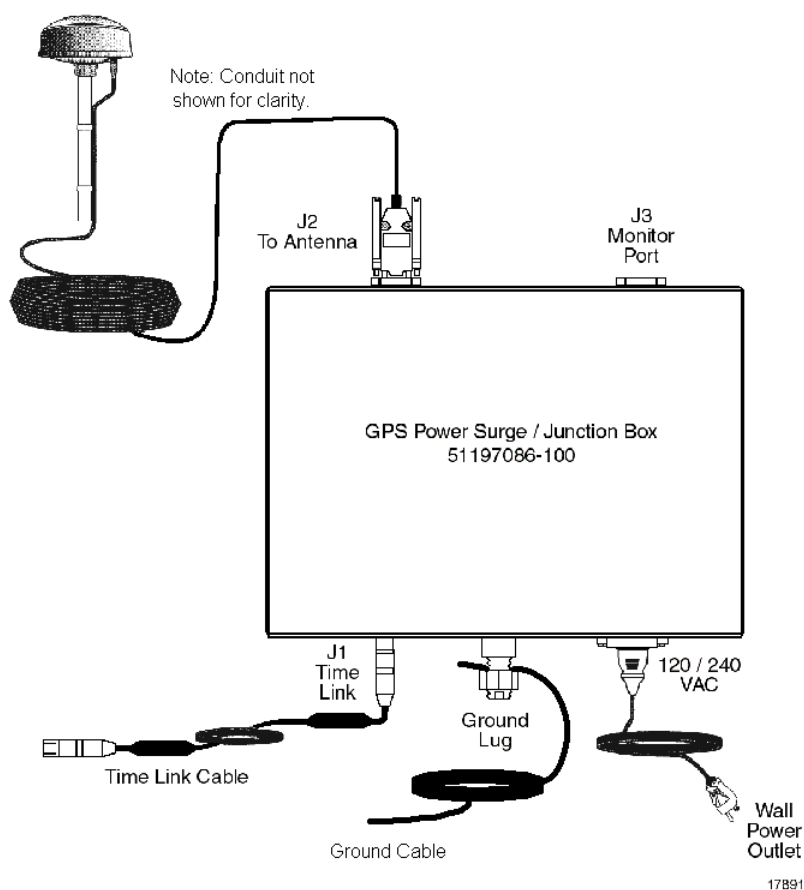


Figure 3-5 GPS Power Surge/Junction Box Cabling Diagram

3. GPS / Time Link Component Descriptions

3.4. Power/Surge Junction Box

GPS Power Surge/Junction Box Function Block

The GPS Power Surge/Junction Box provides dc power to the GPS Antenna, ground connection for the antenna, and drives the Time Link signal distribution.

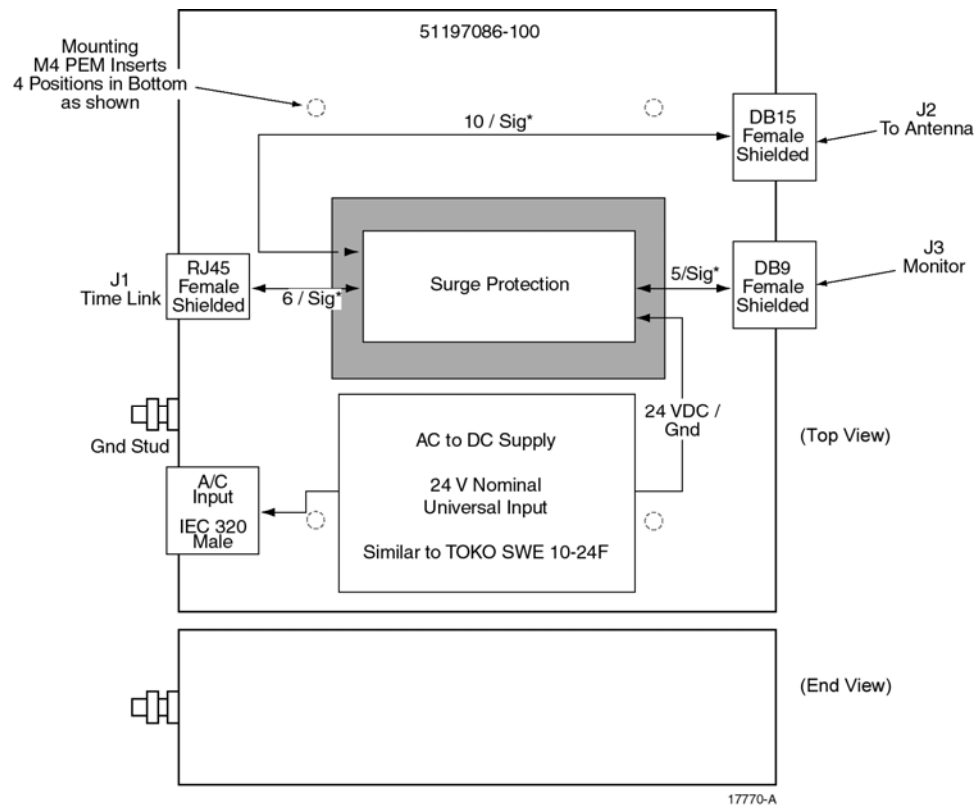


Figure 3-6 GPS Power Surge/Junction Box Block Diagram

Notes:

1. Signals are RS422 levels, low frequency (9600 Baud)
2. DB15 connector end goes to GPS antenna, RJ45 end goes to the computer
3. DB9 connector end is a monitor port used by service
4. A mounting Bracket is required for wall mounting

3. GPS / Time Link Component Descriptions

3.4. Power/Surge Junction Box



ATTENTION

The J3 Monitor Port cabling is reserved for service personnel to check out the setup using a PC.

SAFETY NOTES :



WARNING!

A qualified electrician should perform installation of this product.

To protect the antenna cable, it shall be run through EMT or other suitable metal tubing (see note 1).

To safely and suitably conduct electrostatic transients back to earth, an AWG #6 solid copper wire is required between the 'ground stud' and the protective earth ground grid (e.g., building steel) or individual electrode (NEC 250-83, etc). See note 2.

1. The antenna cable shall be run through EMT conduit or suitable other metal tubing, that is compliant with accepted local electric codes, and shall be suitably bonded to the grounded GPS Power Junction Box by means of the mounting bracket (the antenna housing is non-conductive and can not be bonded).
2. Installation and grounding of the antenna mast and the GPS as shown in this document is only a guide. This serves to illustrate compliance with the INTENT of applicable sections of the electric codes embraced by local electrical authorities having jurisdiction. For domestic, refer to NEC (especially 810-21) and for Canada, refer to CEC Part 1 section 10.
3. Local authorities may revise this installation guide.
4. Per the electric codes embraced in your geographic area, installation of a lightning arrester (such that the antenna is within the curtain of the arrester) may be required.

GPS Power Surge/Junction Box Mounting Bracket

A mounting bracket is required for wall mounting the GPS Power Surge/Junction Box. This is illustrated in "Figure 13: GPS Power Surge/Junction Box Mounting Bracket"

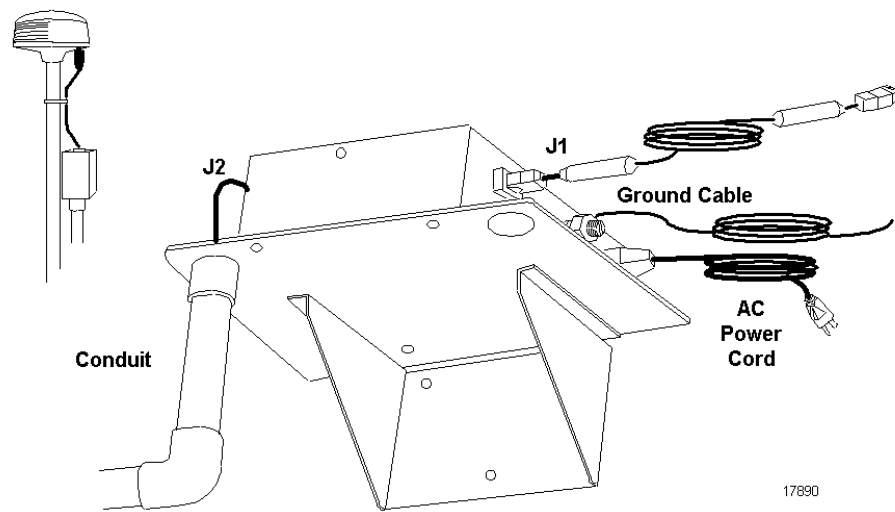


Figure 3-7 GPS Power Surge/Junction Box Mounting Bracket

3. GPS / Time Link Component Descriptions

3.4. Power/Surge Junction Box

Signal and Connector Wiring

The detailed pinout of each cable inter-connection is found in the following table.

Table 3-4 GPS Power Surge/Junction Box Signal Wiring Pinout

RJ45 J1 Pin #	Signal Description	DB15 J2 Pin #	Signal Description	DB9 J3 Pin #
1	N.C. (No Connection)	3	Port B Tx -	2
- - -	N.C. (No Connection)	10	Port B Tx+	6
2	GPS_DETECT	1	N.C. (No Connection)	1
3	1PPS+	4	N.C. (No Connection)	4
4	TSIP-	2	N.C. (No Connection)	8
5	TSIP+	7	N.C. (No Connection)	- - -
6	1PPS-	6	N.C. (No Connection)	- - -
7	N.C. (No Connection)	8	Port B Rcv -	3
- - -	N.C. (No Connection)	11	Port B Rcv +	7
8	Signal Ground / Power Common	5	Signal Ground Common	5
- - -	Power (+24 VDC)	9	N.C. (No Connection)	9
- - -	N.C. (No Connection)	12	N.C. (No Connection)	- - -
- - -	N.C. (No Connection)	13	N.C. (No Connection)	- - -
- - -	N.C. (No Connection)	14	N.C. (No Connection)	- - -
- - -	N.C. (No Connection)	15	N.C. (No Connection)	- - -
Shield	Signal Ground / Case (Pwr common)	Shield		Shield

4. Implementing the GPS and Time Link Options

4.1 GPS Clock Sources and the NCF

General

The user should be familiar with the introductory section "System Timing Requirements vs. Clock Options" prior to reading this material. The purpose of this section is to describe the steps associated with adding GPS clock sources to a system. The addition of a GPS timing reference to a system indicates that the user has a) a desire to provide long term accuracy for the system clock or b) a requirement for system time that has a deterministic relationship to UTC. The assumption is that the system clock sources are to be re-assigned from non-GPS clock sources to clock sources that have capability of referencing the GPS timing. In other words, this is a system upgrade to support a GPS reference.

This section specifically deals with the transfer of the clock source assignments. The "GPS Power Surge/Junction Box Cabling" section describes how the nodes (TPN Bridge or GUS) are physically connected to the GPS antenna through the power / junction box, so those details are not repeated here.

Since the TPNB and LCNP4 with GPS capability do not support analog clock generation, all nodes must contain KxLCN or LCNPx kernels. If this is not the case, the kernel board updates should be performed prior to insertion of GPS clock sources.

The Network Configuration File (NCF)

The Network Configuration File (NCF) contains the node addresses of the clock sources. Two nodes are assigned to act as clock sources. The system software is responsible for the active role performed by each of the two-clock source nodes. The role of Master clock source and Slave clock source is determined by System Clock Driver software. This requires no user intervention. The only function this user is required to perform is the selection of the nodes that are to act as clock sources.

The De-Activation of old Clock Sources / Activation of new Clock Sources

The sequence for de-activation of the current clock source nodes and activation of the new clock source nodes is described in the following two procedures. The difference between the two procedures has to do with which node types are to become the new clock sources.

4. Implementing the GPS and Time Link Options

4.1. GPS Clock Sources and the NCF

De-Activation of Clock Sources and Activation of GUS w/GPS Clock Sources

This first procedure describes the traditional sequence for RNOS based clock sources (i.e., GUS nodes w/ GPS are to become the new clock sources). The text in **COURIER** is information that appears on the CRT in response to the completion of the steps preceding the “on-screen” instructions.

Step	Action
1	From the Engineering Menu, select SYSTEM WIDE VALUES .
2	Select CLOCK SOURCES
3	Enter the new Clock Source Node Addresses (lowest number in the left port). (Example: 10 12) Press ENTER .
4	Select F1 (Check) and follow the instructions shown on the monitor. (Instructions also duplicated below.) 1. Install changes in NCF (F2) <i>The following steps outline the procedure for loading the changes into the network. You may install other changes before performing these steps.</i> 2. For each node which is the old clock source: 6 7 a) <i>SHUTDOWN the node.</i> b) <i>LOAD the node using the new NCF.</i> 3. For each node which is the new clock source. 10 12 c) <i>SHUTDOWN the node.</i> d) <i>LOAD the node using the new NCF.</i> 4. For EACH node not already re-loaded. a) <i>SHUTDOWN the node.</i> b) <i>LOAD the node using the new NCF.</i>

4. Implementing the GPS and Time Link Options

4.1. GPS Clock Sources and the NCF

De-Activation of Clock Sources and Activation of TPN Bridge Clock Sources

This procedure provides the sequence for the transition from RNOS based clock sources (i.e., US, NIM, GUS) to TPN Bridge clock sources.

Step	Action
1	From the Engineering Menu, select SYSTEM WIDE VALUES.
2	Select CLOCK SOURCES
3	Enter the new Clock Source Node Addresses (lowest number in the left port). (Example: 95 96) Press ENTER.
4	Select F1 (Check) and follow the instructions shown on the monitor. (Instructions also duplicated below.) 1. Install changes in NCF (F2) <i>The following steps outline the procedure for loading the changes into the network. You may install other changes before performing these steps.</i> 2. For each node which is the old clock source: 7 7 a) SHUTDOWN the node. b) LOAD the node using the new NCF.
6	Perform the insertion of Bridge Nodes as described in "TPN Bridge Hot Insertion" section contained in the <i>TPN Bridge Overview and Implementation Guide</i> .
7	1. For <u>each</u> node not already re-loaded. a) SHUTDOWN the node. b) LOAD the node using the new NCF.

4. Implementing the GPS and Time Link Options

4.1. GPS Clock Sources and the NCF

Verification of Clock Synchronization Status

After the new clock sources are activated, the user may utilize the “CLOKSTAT” schematic to view the current synchronization state of each node.

For RNOS based clock sources (GUS nodes) the schematic should show MASTER and SLAVE mode assignments to those clock sources with all other nodes indicating LISTENER state with DIGSYNCH sync status.

For TPN Bridge based clock sources, the Bridge nodes do not appear in the list, but all other nodes should be in the LISTENER state with DIGSYNCH sync status.

4.2 Installing and Verifying the GPS / Time Link

General

The following procedures are intended to guide a user through the installation and check out of GPS / Time Link for TPN Bridge or GUS / APP nodes or Time Link without GPS for TPN Bridge Applications. For Time Link without GPS (not applicable to GUS / APP), the Installing the GPS and Time Link" procedure is skipped. In this case go directly to the "Checking out the GPS and Time Link – TPN Bridge Applications" procedure.

Each procedure is presented in a logical sequence of steps with a place to mark the completion of that step. In general, the steps should be followed in sequence for the best results. Towards the end of the Checkout process for both the TPN Bridge application and GUS / APP node application some intentional failure conditions are introduced along with expected results and recovery process. These steps are intended to provide a more thorough checkout of the time sync functionality. They may be skipped if it is not desirable to introduce these intentional faults.

Installing the GPS and Time Link

Step	Installation Action	Done 3
1	Choose a suitable site for installation of the GPS antenna. See Figure 3-2 and "Antenna placement" in Section 3.1 for criteria regarding antenna placement. For best performance, this location should be as far as possible from transmitting antennas and should have a relatively unobstructed view of the sky.	☐
2	Determine the desired position for the GPS power / junction box. This location must provide an earth ground (building steel) connection point the GPS power / junction box. To safely and suitably conduct electrostatic transients back to earth, an AWG #6 copper wire is required. This wire should be as short as possible.	☐
3	Mount the GPS power / junction box in the desired location. Wall mount bracket 51305574-100 is recommended for attachment to a wall surface. Install the ground cable. Do not connect the ac power at this time.	☐
4	Install a 1" –14 marine and UNS threaded pipe antenna mast for mounting the GPS antenna in the location identified in step 1. Secure the mast to a solid structure so that it is oriented vertically.	☐

4. Implementing the GPS and Time Link Options

4.2. Installing and Verifying the GPS / Time Link

Step	Installation Action	Done 3
5	Determine routing path between the antenna and junction box and route the GPS antenna cable from the power / junction box to the antenna mast. Check local electric codes for any special requirements that may apply in the installation of this cable. Install the GPS antenna cable and connect the DB15 end to the power / junction box. Avoid sharp bends, kinks and sharp or abrasive surfaces when routing the cable.	☐
6	Install the GPS antenna on the 1"-14 marine and UNS threaded pipe mast. Thread the antenna onto the mast and hand-tighten until snug. Caution: Do not over-tighten the antenna on the mast. This could damage the threaded socket in the base of the antenna. In addition, do not use thread-locking compounds, since they can corrode plastic.	☐
7	Connect the GPS antenna cable to the 12-pin connector. The locking ring on the GPS antenna cable should lock the cable securely to the antenna when it is rotated. Upon completion of the initial checkout of the GPS antenna, it is recommended that waterproofing tape be applied to the 12-pin connector, the GPS antenna cable be secured to the mast with tie-wraps, and any required strain relief be added at the power junction box.	☐
8	Route the Time Link cables between the power / junction box and the appropriate TPN Bridge nodes OR GUS / APP Nodes (See GUS / APP Applications) using Time Link cables (51305562-4xx where xx is cable length), T-connectors (51192708-100) and a Terminator (51192707-100) at the end of the daisy chain. (See Figure 1-1 in this guide for a diagram showing how these elements are used). Connect all of the Time Link components together, but do not attach the short 0.5-meter cables into the Time Link ports of the TPNB PWAs at this time.	☐
9	Connect the source end of the Time Link cable daisy chain to the power / junction box and connect the power junction box to the ac power.	☐

4. Implementing the GPS and Time Link Options

4.2. Installing and Verifying the GPS / Time Link

Checking out the GPS and Time Link – TPN Bridge Applications

Step	Checkout Action	Done 3
1	The following process is recommended to provide the best method of identification of any problem in the new Time Link cables, T-connectors, junction box and antenna. This calls for checking the proper operation of the Time Link cables, at the TPN Bridge locations, prior to the addition of the GPS source to the Time Link cable system. Check to see that all of the Time Link components are connected together and that the short 0.5-meter cables are not yet connected to the TPNB PWAs. Check to see that a Terminator (51192707-100) is installed at the far end of the daisy chain.	☐
2	Disconnect the Time Link cable segment that goes to the power junction box at the T-connector (1 st in the daisy chain) and replace it temporarily with a Terminator (51192707-100).	☐
3	Connect the open ends of the 0.5-meter Time Link cables to the appropriate Time Link Port (TLI) connectors on the TPNB PWAs.	☐
4	Look for the One Pulse Per Second (1PPS) flashing LED on all connected TPNBs. The presence of the 1PPS LED illumination on a 1 second periodic basis is an indication that at least one half of the 2 signal pairs (1PPS , TSIP) is properly connected from the source (in this case the Primary TPN Bridge which is configured as TPN Master) If the 1PPS LEDs are not flashing on some or all of the connected TPNBs, then the problem may be a bad connection between the Primary TPN Master and the other TPN Bridge Nodes. Identify and correct this problem before proceeding.	☐
5	Verify the Progress State in the TPN Bridge View Status display to be 255 for each TPN Bridge and that the "Time Link Port Disabled" status box is NOT checked. Verify that "Init Done" status box is checked and the "Am Synched" status box is checked.	☐
6	Verify that the system time is approximately correct. The current system time will be used to determine the zone offset to be applied to UTC. When the GPS signal is added to the Time Link cable, the closest ½ hour will be selected as the Zone offset until a new system time is set.	☐
7	Verify using the CLOKSTAT schematic that all nodes are properly synchronized. If all status reports are as expected, then the portion of the Time Link cable that is connected to the TPN Bridge nodes is functional.	☐

4. Implementing the GPS and Time Link Options

4.2. Installing and Verifying the GPS / Time Link

Step	Checkout Action	Done 3
8	Next is the insertion of the GPS antenna signal to the Time Link cable. Check to make sure that ac power has been applied to the power / junction box and that the GPS antenna cable and Time Link cables are connected at the junction box. NOTE: When power is first applied to the Smart antenna, it requires several minutes from a cold start to acquire and track the GPS satellite signals. During this period of time no timing data will be sent from the antenna. Therefore, be sure to provide adequate time from the application of antenna power before checking to see that the 1PPS and TSIP packets are being generated. When adequate time has been allowed for the Smart GPS antenna to acquire and track the satellite signals, remove the Terminator that was temporarily inserted at the 1st Time Link T-connector and insert the Time Link cable from the power junction box.	☐
9	Check to be sure that all of the 1PPS LED's are still flashing. If no LEDs are flashing on the TPNB pwa's, then the problem may be: 1. Lack of power to the antenna. 2. Absence of valid satellite signal on at least 3 of the 8 channels. 3. A bad connection between the GPS antenna and the power junction box. 4. A bad connection between the power junction box and the first T-connector. Disconnect the Time Link Cable from the 1st T-connector, replace the Terminator and determine the cause of the problem before proceeding. Note: The Monitor port on the power junction box may be connected to a COM port on a PC by means of an RS422-RS232 converter to monitor the health of the antenna. This may be useful in the identification of the source of trouble in this step. If all of the TPNB 1PPS LEDs are flashing, proceed to the next step.	☐

4. Implementing the GPS and Time Link Options

4.2. Installing and Verifying the GPS / Time Link

Step	Checkout Action	Done 3
10	Verify the Progress State in the TPN Bridge View Status display to be 255 for each TPN Bridge and that the "Time Link Port Disabled" status box is NOT checked. Verify that "Init Done" status box is checked and the "Am Synched" status box is checked If these conditions are not true, disconnect the GPS antenna from the Time Link cable at the 1st T-connector and diagnose the problem. NOTE: When the GPS signals are not functional, a two-minute periodic – "Re-Sync" status will be observed while the GPS is connected into the Time Link cables. If the status is as expected, proceed to next step.	☐
11	Verify using the CLOKSTAT schematic that all nodes are properly synchronized. If all status reports are as expected, then the portion of the Time Link cable that is connected to the TPN Bridge nodes is functional.	☐
12	Verify that the system time is what is expected. Any error in the system time should now be eliminated and all time sets performed while referenced to the GPS will set the system time to the nearest ½ hour offset from UTC.	☐
13	If additional time sets are allowed to be performed, set the time to some other values and observe the effect of performing a time set with several minutes of error in the time set value.	☐
14	Perform a fail-over from Primary to Secondary and check to make sure that the TPN Mastership function transitions to the active primary and the resting state following transition is the same as prior to the fail-over.	☐
15	Remove ac power to the junction box and monitor the two-minute periodic re-sync attempts. Restore power and check to make sure the resting state following recovery of the GPS signals is the same as prior to the induced power outage.	☐
16	Perform any remaining cable dressing (cable ties, strain relief, etc.)	☐
17	Continue to monitor the system synchronization status and time value for a period of no less than a week.	☐
18	Checkout is complete.	☐

4. Implementing the GPS and Time Link Options

4.2. Installing and Verifying the GPS / Time Link

Checking out the GPS and Time Link – GUS / APP Node Applications

Step	Checkout Action	Done 3
1	The following process is recommended to provide the best method of identification of any problem in the new Time Link cables, T-connectors, junction box and antenna in the GUS / APP Node application. In the GUS / APP Node application, there is no method of splitting the checkout into two pieces as is done in the TPN Bridge application. Since the only source of Time Link Data and 1PPS signals in the GUS / APP node application is the GPS antenna, the Time Link components can not be tested separately from the portion leading up to the GPS antenna. Check to see that all of the Time Link components are connected together and that the short 0.5-meter cables are not yet connected to the LCNP4-GPS PWAs. Check to see that a Terminator (51192707-100) is installed at the far end of the daisy chain.	☐
2	Check the system synchronization status using the CLOKSTAT schematic to ensure that everything is in a normal operating condition before proceeding.	☐
3	Verify that the system time is approximately correct. The current system time will be used to determine the zone offset to be applied to UTC. When the GPS signal is added to the Time Link cable, the closest ½ hour will be selected as the Zone offset until a new system time is set.	☐
4	Check to make sure that ac power has been applied to the power / junction box and that the GPS antenna cable and Time Link cables are connected at the junction box. NOTE: When power is first applied to the Smart antenna, it requires several minutes from a cold start to acquire and track the GPS satellite signals. During this period of time no timing data will be sent from the antenna. Therefore, be sure to provide adequate time from the application of antenna power before checking to see that the 1PPS and TSIP packets are being generated.	☐
5	When adequate time has been allowed for the Smart GPS antenna to acquire and track the satellite signals, connect the open ends of the 0.5-meter Time Link cables to the appropriate Time Link Port (TLI) connectors on the LCNP4-GPS PWAs.	☐

4. Implementing the GPS and Time Link Options

4.2. Installing and Verifying the GPS / Time Link

Step	Checkout Action	Done 3
6	Look for the One Pulse Per Second (1PPS) flashing LED on the Master Clock Source connected LCNP4-GPS. The presence of the 1PPS LED illumination on a 1 second periodic basis is an indication that at least one half of the 2 signal pairs (1PPS , TSIP) is properly connected from the source (in this case the GPS antenna.) If the LED is not flashing on the Master Clock LCNP4-GPS, then the problem may be: 1. Lack of power to the antenna. 2. Absence of valid satellite signal on at least 3 of the 8 channels. 3. A bad connection between the GPS antenna and the power junction box. 4. A bad connection between the power junction box and the first T-connector. 5. A bad Time Link Cable. 6. A bad T-connector. Determine the cause of the problem before proceeding. Note: The Monitor port on the power junction box may be connected to a COM port on a PC by means of an RS422-RS232 converter to monitor the health of the antenna. This may be useful in the identification of the source of trouble in this step. If the Master Clock LCNP4-GPS 1PPS LED is flashing, proceed to the next step.	☐
7	Verify using the CLOKSTAT schematic that all nodes are properly synchronized. If all status reports are as expected, then the GPS antenna, power junction box, the Time Link cable components and Time Link port on the LCNP4-GPS is functional. If the Master clock status is not as expected, then the problem is likely related to a problem in the TSIP packet signal path. If okay, proceed to the next step.	☐
8	Verify that the system time is what is expected. Any error in the system time should now be eliminated and all time sets performed while referenced to the GPS will set the system time to the nearest ½ hour offset from UTC.	☐

4. Implementing the GPS and Time Link Options

4.2. Installing and Verifying the GPS / Time Link

Step	Checkout Action	Done 3
9	If additional time sets are allowed to be performed, set the time to some other values and observe the effect of performing a time set with several minutes of error in the time set value.	☐
10	Perform a fail-over of the Master Clock and check to make sure that the Slave assumes role of Clock Master. Check to make sure that the system synchronization status returns to normal after the transition of the Slave to Master.	☐
11	Remove ac power to the junction box and make sure that the Master clock is able to recover after restoration of the GPS sync reference.	☐
12	Perform any remaining cable dressing (cable ties, strain relief, etc.)	☐
13	Continue to monitor the system synchronization status and time value for a period of no less than a week.	☐
14	Checkout is complete.	☐

5. Clock Status Displays

5.1 CLOCKSTAT Schematic

Description

The “CLOCKSTAT” schematic, as shown in Figure 5-1, is used extensively to determine the overall Synchronization state of the TPS nodes. Each node is listed in sequence of lowest to highest node address along with its Mode of operation (LOCAL, LISTENER, SLAVE and MASTER), cable selection status (normally ALTCBLA or ALTCBLB), synchronization state (PRECREF, DIGSYNCH, SUBSYNCH, NOSYNC) and translator status (NONTR, TRNSLTR).

CLK_MOD	CLK_CBL	CLK_SYN	CLK_TRN	Node Address	CLK_MOD	CLK_CBL	CLK_SYN	CLK_TRN	Node Address
LISTENER	ALTCBLB	DIGSYNCH	NONTR	ND001	SLAVE	ALTCBLB	DIGSYNCH	NONTR	ND49
MASTER	ALTCBLA	PRECREF	NONTR	ND004					ND50
LISTENER	ALTCBLA	DIGSYNCH	NONTR	ND005					ND51
				ND007					ND52
				ND008					ND53
				ND009					ND54
				ND010					ND55
				ND011					ND56
				ND012					ND57
				ND013					ND58
				ND014					ND59
				ND015					ND60
				ND016					ND61
				ND017					ND62
				ND018					ND63
				ND019					ND64
				ND020					ND65
				ND021					ND66
				ND022					ND67
				ND023					ND68
				ND024					ND69
				ND025					ND70
				ND026					ND71
				ND027					ND72
				ND028					ND73
				ND029					ND74
				ND030					ND75
				ND031					ND76
				ND032					ND77
				ND033					ND78
				ND034					ND79
				ND035					ND80
				ND036					ND81
				ND037					ND82
				ND038					ND83
				ND039					ND84
				ND040					ND85
				ND041					ND86
				ND042					ND87
LISTENER	ALTCBLA	DIGSYNCH	NONTR	ND043					ND88
				ND044					ND89
				ND045					ND90
				ND046					ND91
				ND047					ND92
				ND048					ND93
									ND94
									ND95
									ND96

Figure 5-1 CLOCKSTAT Schematic Display

5. Clock Status Displays

5.1. CLOCKSTAT Schematic

Interpreting the schematic

The training module publication, *Interpret System Clock Functions* (L61524 LCN), provides a very detailed explanation of the fields listed above and that level of detail is not repeated here. The most important column is left most position. The user should look for a single node as MASTER, a single node as SLAVE and all other nodes as LISTENERS *except* in specific TPN Bridge applications where the TPN Bridge nodes have been configured as both TPN and Segment Masters. In this case, the NCF assignments will show the node addresses of the Bridge Nodes as clock sources. Since the bridge nodes do not participate *directly* in the reporting associated with the updates of the CLOKSTAT schematic, these nodes do not show up as MASTER and SLAVE. However, when the system is operating in the expected manner, all other nodes should still report as LISTENER.

5.2 TDC Status Display

Introduction

The application that generates this display is installed on a TPN Bridge node. This application is called the TPN Bridge Viewer. When invoked, you choose the “View Bridge Status” button to get the “TDC Status” display as shown in Figure 5-2.

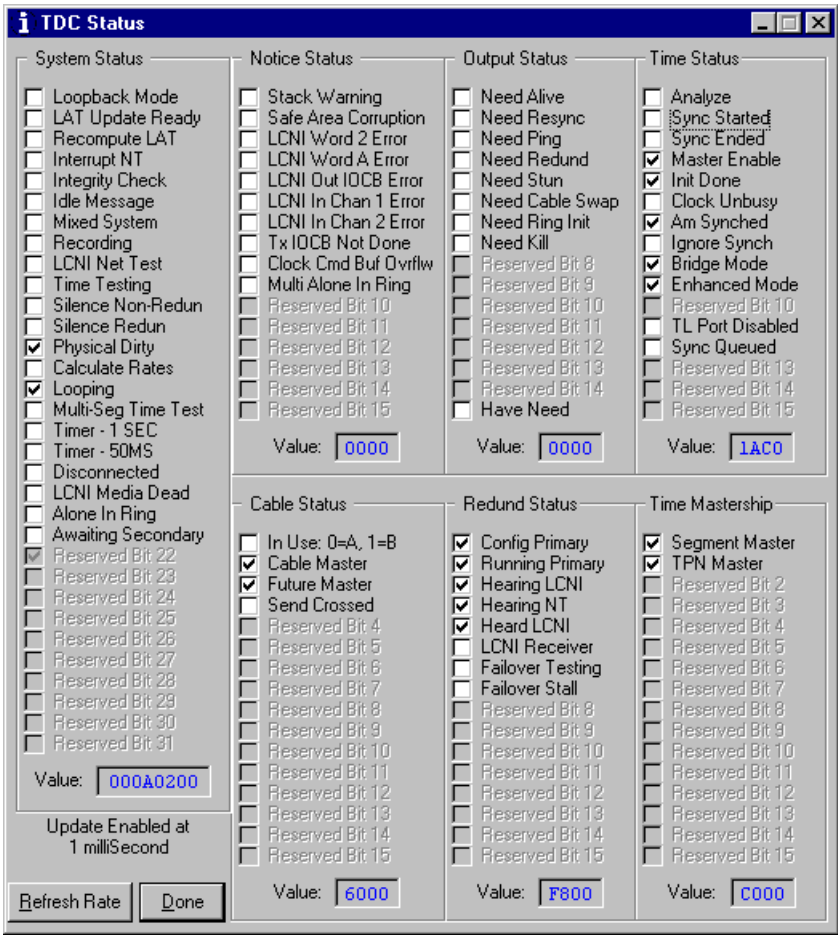


Figure 5-2 TDC Status Display (TPN Bridge)

5. Clock Status Displays

5.2. TDC Status Display

Status display description

The TDC Status Display (TPN Bridge Nodes Only) contains information regarding the Synchronization state of the particular TPN Bridge node on which the display is presented. This display provides a method of determining the health of TPN Bridge time synchronization from the perspective of a single Bridge node. The important fields in this display (regarding Time Synchronization) are presented in the columns identified as “Time Status” and “Time Mastership.”

Table 5-1 shows the items that are checked in the Time Status entity for normal conditions as shown in TDC Status dialog page (see Figure 5-2).

Table 5-1 TPN Bridge Time Status Check Box Status (Normal Conditions)

	TPN and Segment Master Primary	TPN and Segment Master Secondary	TPN Master Primary	TPN Master Secondary	Segment Master Primary / Secondary
Analyze					
Sync Started					
Sync Ended					
Master Enable	X		X		
Init Done	X	X	X	X	X
Clock Unbusy					
Am Synched	X	X	X	X	X
Ignore Sync					
Bridge Mode	X	X	X	X	X
Enhanced Mode	X	X	X	X	X
TL Port Disabled					
Sync Queued					

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